

When Is Surrogate-Assisted Multiobjective Ensemble Weighting Trustworthy?

A Replicated Study of Mixture Designs, Scheffé Surfaces and Normal Boundary Intersection on the Classifier-Weight Simplex

Caio Tertuliano Ribeiro* Matheus Costa Pereira Anderson Paulo de Paiva

Universidade Federal de Itajubá, Itajubá, MG, Brazil

Abstract

Choosing soft-voting ensemble weights is a mixture experiment on a simplex, already addressed by an established mixture-design, response-surface and Normal Boundary Intersection pipeline. We ask not whether that pipeline yields a Pareto front, but when the front can be trusted. Three arms defined by two methodological factors — the source of the objectives and the source of the anchors — run over four public binary tabular datasets $\times$ 30 partitions, paired and overlap-corrected, behind controls the framework lacks: a pre-specified external reliability gate; revalidation of every candidate on the real objectives; an empirical reference whose sampled core is constructed independently of the surrogate and then augmented with all candidate sets; and an untouched holdout. Surrogate adequacy is dataset- and metric-dependent: ROC-AUC surfaces pass the gate in 0, 5, 16 and 30 of 30 partitions, log-loss in 30, 3, 12 and 30. Among failures attributable to the surrogate pipeline, anchor misplacement has the largest observed effect: real anchors improve both primary endpoints in all 30 partitions of the two datasets whose surfaces fail, in 23–24 of 30 on Porto Seguro and 24 of 30 on UCI credit. A control injecting only those anchors attributes most of that gap to the returned extreme solutions on the first two, and to the relocated construction on UCI credit, where subproblem convergence, not surrogate error, dominates. The gate detects unusable surfaces but not misplaced anchors. Continuous and step deployment costs disagree about which of six candidate sets wins in 10 to 24 of 30 partitions; at a matched budget NSGA-II beats the metamodel-free arm. The diagnosis concerns one-shot, no-infill pipelines; four datasets give a four-point cross-dataset pattern, not a tested claim.

Keywords: ensemble weighting; mixture design of experiments; Scheffé canonical polynomials; response surface methodology; normal boundary intersection; surrogate-assisted multiobjective optimization; inference cost; replicated evaluation.

1 Introduction

A weighted average of several classifiers’ predicted probabilities is among the oldest and most reliable devices in applied machine learning (Hansen and Salamon, 1990; Wolpert, 1992; Kittler et al., 1998; Caruana et al., 2004). Its weights are not free parameters: to keep the output a probability they must be non-negative and sum to one, and non-negativity with normalization is what makes stacked combinations stable (Breiman, 1996). The feasible set is the unit simplex, so ensemble weighting is a *compositional* optimization problem.

That has a methodological consequence: optimization over a simplex is the subject of mixture experiments, a mature branch of experimental design (Scheffé, 1958, 1963; Cornell, 2002). The classifiers are the components, a pure vertex is a single model, the centroid uniform soft

*Corresponding author. caio.tertu99@gmail.com

voting, and a Scheffé polynomial the natural response surface. The correspondence is exact and already exploited for portfolio weights (Mendes et al., 2016; Leal et al., 2022) and neural-network ensembles (Moreira et al., 2021), and combined with Normal Boundary Intersection for forecast combination (Bacci et al., 2019), load forecasting (Rocha et al., 2025) and multiobjective engineering optimization (Pereira et al., 2025). Most directly, Kwon et al. (2024) treat five heterogeneous *classifiers*’ weights as mixture components and maximize a Scheffé polynomial of accuracy or F_1 on eleven tabular binary datasets.

We therefore claim no part of that construction: mixture design over combination weights, a Scheffé surrogate over the simplex, and NBI on that surrogate are established prior work, individually and for classifier ensembles. What is not established is whether the construction stays *trustworthy* once the problem becomes genuinely multiobjective — this paper’s question.

1.1 Why the transfer is not routine

The applications in which this framework matured share three properties that classifier ensembles do not.

First, *the responses are smooth*: tool life or process cost varies smoothly with cutting speed, where a low-order polynomial is a defensible local model. The area under the ROC curve is instead a rank statistic, piecewise constant in the weights. No experimental design repairs a misspecified model class, and a smooth surrogate fitted to a step-like response misrepresents it systematically.

Second, *the decision variables are the mixture*. When the simplex holds scalarization weights, as in the lineage above, NBI’s anchors are optima of the surrogate in another space; when it holds the decision variables themselves, the anchors are feasible-region points — often vertices — so an anchor error displaces the geometry in which every subproblem is posed. NBI derives its whole search geometry from the payoff matrix of individual optima (Das and Dennis, 1998), so errors there propagate globally. That misplaced anchors damage the construction is established: payoff-table minima can differ substantially from the true minimum criterion values over the efficient set (Isermann and Steuer, 1988), the nadir point is hard to estimate (Deb et al., 2010), and the choice of the individual minima has itself been treated as a design lever (Herrmann-Wicklmayr and Fließkamp, 2026). What that literature, and the normalization remedies built on it (Messac et al., 2003), asks about is estimating the *true* objectives; none of it is about a surrogate standing in for them.

Third, *one objective is a step function*: a model with any non-negligible weight must be loaded and executed, one with none costs nothing, so deployment cost is not proportional to a weight. A method requiring differentiable objectives can only optimize a continuous relaxation, with no guarantee that it and the quantity actually paid agree about which solution is best.

Each breaks a different part of the pipeline — the model class, the search geometry, the objective definition — and none is visible in the aggregate fit statistics such studies conventionally report.

1.2 Approach

We decompose the framework. Because the ensemble output is linear in the weights and base-model probabilities can be cached, evaluating a weight vector on the true objectives costs one matrix–vector product (0.4 to 2.1 milliseconds here) — cheap enough to run the *same* NBI three times per replication, changing only where objectives and anchors come from.

- **NBI-A** optimizes the Scheffé surfaces with anchors from their own optima: the inherited construction and our control.

- **NBI-B** optimizes the same surfaces with anchors from the real single-objective optima; A versus B isolates *anchor misplacement*.
- **NBI-C** drops the metamodel and runs the same NBI on the exact out-of-fold objectives — a metamodel-free NBI in the published sense of Gellerich and Majschak (2023), whose subproblems are physical experiments; B versus C isolates residual *interior surrogate error*.

All three arms fit their surfaces once from a fixed design and then optimize. This is the *sequential* form of surrogate-assisted optimization (Tabatabaei et al., 2015) — one-shot, with no infill — and it is the form the lineage above uses; nothing here tests model-managed, infill-based surrogate-assisted optimizers such as ParEGO, MOEA/D-EGO or SMS-EGO (Knowles, 2006; Zhang et al., 2010; Ponweiser et al., 2008), which evaluate new points on the true objectives inside the search loop and exist precisely to repair the kind of surrogate error diagnosed here. Every conclusion in this paper is scoped to the one-shot, no-infill case (Section 2.5).

We also run NSGA-II on the same objectives at a matched real-objective evaluation budget. Four controls the lineage does not contain surround the decomposition: external validation of every surface on 100 unseen compositions, with a reliability gate fixed before the replicated benchmark; revalidation of every candidate on the true objectives before any indicator; an empirical Pareto reference whose sampled core of at least 10^5 points per replication is built independently of the surrogate and checked for convergence, then augmented with every candidate set for scoring; and an untouched holdout never used for fitting, weighting, thresholding or selection.

Four public binary tabular datasets — customer-transaction prediction (Santander), insurance-claim processing (BNP Paribas Cardif), auto-insurance claim prediction (Porto Seguro) and consumer-credit default (UCI credit) — are partitioned 30 times each, giving 120 replications with recorded seeds. The four prediction tasks differ, but all four come from financial-services applications, so any cross-dataset statement is made within a single application domain. Because partitions of one dataset overlap heavily, we treat the replication distribution as partition sensitivity, not a sample from a population: comparisons are paired by partition and tested with the Nadeau–Bengio correction (Nadeau and Bengio, 2003), the dataset is the unit of generalization, and nothing is pooled across the 120 replications.

1.3 Research questions

- RQ1** How accurately do Scheffé surfaces predict ensemble performance on compositions they have not seen, and does this depend on the dataset, on the metric, or on both?
- RQ2** How stable are the fitted surfaces across resampling, and does the classical mixture-design reading of the interaction coefficients survive confrontation with the real objectives?
- RQ3** What goes wrong when NBI is anchored on surrogate optima?
- RQ4** Does replacing surrogate anchors with real ones repair the front, and does the reliability gate predict when this is necessary?
- RQ5** How does surrogate-assisted NBI compare with metamodel-free NBI, what does the difference cost, and how do both compare with a canonical evolutionary optimizer at a matched evaluation budget?
- RQ6** How large is the actual conflict between discrimination and calibration?
- RQ7** Does the definition of deployment cost change which method wins?
- RQ8** Do solutions selected on out-of-fold data transfer to an untouched holdout?
- RQ9** Would a third of the replications have produced the same conclusions?

1.4 Contributions

1. **A controlled decomposition of a one-shot surrogate-assisted multiobjective pipeline into surrogate error, anchor misplacement and solver behaviour**, through three NBI variants defined by two methodological factors. The ingredients are not new and are not claimed: anchor and ideal/nadir misestimation is an established failure mode (Isermann and Steuer, 1988; Deb et al., 2010; He et al., 2021), the choice of individual minima has already been treated as a design lever (Herrmann-Wicklmayr and Fläßkamp, 2026), and NBI run metamodel-free on measured objectives with real anchors is published (Gellerich and Majschak, 2023). What we did not find is a study that computes the anchors of a CHIM method both from a fitted surrogate and from the true objectives on the same problem and compares the resulting fronts, so what is new here is the surrogate *provenance* of the anchor error and the controlled comparison of the two constructions; the nearest ensemble-weighting precedent (Kwon et al., 2024) optimizes one metric at a time and has no anchors or front.
2. **Evidence that, among failures attributable to the surrogate pipeline, anchor misplacement has the largest observed effect.** Real anchors improve both primary endpoints in 30 of 30 partitions on the two datasets whose ROC-AUC surfaces almost never pass the gate, in 23 (IGD⁺) and 24 (hypervolume) of 30 on Porto Seguro, and in 24 of 30 on UCI credit. Removing the surrogate entirely, once anchors are correct, adds much less — except on UCI credit, where subproblem convergence rather than surrogate error dominates, the counterexample that prevents a universal claim.
3. **A demonstration that an external reliability gate detects unusable surfaces but not misplaced anchors.** On one dataset the two properties diverge: the partitions where surrogate-anchored NBI collapses are precisely those where the parsimony rule selects a surface that fits held-out compositions *better* and passes the gate more often. Surface quality and argmax quality are different properties, and only the second governs NBI.
4. **A replicated evaluation architecture for ensemble-weight optimization** — external surrogate validation, real-objective revalidation, a reference with a method-independent sampled core and holdout confirmation, over 4 datasets $\times$ 30 partitions with overlap-corrected paired inference. Three findings it overturned — two of them artifacts that survived an earlier ten-partition analysis — are reported explicitly.
5. **Two findings about how these surfaces should be read.** The classical synergism criterion $\beta_{ij} > |\beta_i - \beta_j|$ holds for the leading pair’s fitted surfaces on all four datasets in every partition, yet the real blend beats its better member on only one; across all pairs the surrogate over-predicts blending gains 20 times and under-predicts none. And the continuous cost these methods optimize disagrees with the step cost deployment pays often enough to change the winning method in up to 24 of 30 partitions. That a change of cost accounting can reorder a comparison is known across ensemble and cascade architectures (Wang et al., 2022); what is reported here is the disagreement between two accountings applied to one and the same weight vector.

We claim neither that NBI outperforms other multiobjective optimizers — at a matched evaluation budget NSGA-II outperforms its metamodel-free arm on all four datasets — nor that it produces more uniformly spaced fronts, nor that any method here is a general-purpose recommendation. The contribution is the set of conditions under which an established framework can and cannot be trusted in a new problem class, and the controls needed to tell the difference — stated for the one-shot, no-infill form of that framework, since no arm here performs iterative infill.

2 Related work

2.1 Classifier ensemble weighting

Weighted voting and weighted probability averaging are long established (Hansen and Salamon, 1990; Kittler et al., 1998; Dietterich, 2000; Kuncheva, 2014); the weights are classically set by stacked generalization on out-of-fold predictions (Wolpert, 1992; Breiman, 1996; LeBlanc and Tibshirani, 1996; van der Laan et al., 2007; LeDell et al., 2016), by greedy forward selection with replacement (Caruana et al., 2004), or by numerical optimization (Zhang and Zhou, 2011; Yao et al., 2018; Shahhosseini et al., 2022; Hassan et al., 2026; Xu et al., 2026); Fumera and Roli (2005) gives the theoretical analysis of weighted averaging. Throughout, the simplex is almost always respected as a constraint rather than designed over as an experimental region, and essentially all of this work is single-objective.

2.2 Multiobjective ensemble optimization

Multiobjective formulations of ensemble learning divide by their decision variables: the *members*, evolved under accuracy-versus-diversity objectives (Abbass, 2003; Chandra and Yao, 2006; Jin and Sendhoff, 2008); a *subset*, with binary membership variables (Dos Santos et al., 2008; Qian et al., 2015); or — the family relevant here — *continuous combination weights* under conflicting criteria, optimized by metaheuristics (Ekbal and Saha, 2011, 2013; Bandyopadhyay et al., 2008; Onan et al., 2016; Rezaei et al., 2019; Zhao et al., 2018; Vrugt et al., 2006; Galván et al., 2026). Surveys organize the field by generation, selection and combination (Ribeiro and Reynoso-Meza, 2020).

This family is therefore *not* new, and we do not claim it. Absent from this cluster is any designed experiment and fitted response surface over the weight simplex, any classical scalarization method such as NBI, and — with the partial exception of the sparsity term in Zhao et al. (2018) — any treatment of deployment cost as an objective. Fronts are rarely replicated over resampled partitions with corrected inference (Demšar, 2006).

2.3 Mixture experiments and response surfaces

Mixture experiments address exactly the constraint set of Equation (2), through simplex-lattice designs and the canonical polynomials that carry Scheffé’s name (Scheffé, 1958, 1963; Cornell, 2002; Myers et al., 2016). Because $\sum_i w_i = 1$, mixture-model coefficients are not component effects in the ordinary sense and separate measures are needed to interpret them (Piepel, 1982).

The classical reading of a positive interaction coefficient is *synergistic blending* (Cornell, 2002): a statement relative to the chord, not relative to the better pure component, and the distinction turns out to matter (Section 5.2).

Mixture designs are beginning to appear in machine learning (Karl et al., 2023; Leal et al., 2025; Xie et al., 2023; Ye et al., 2025; Liu et al., 2025; Chen et al., 2025; Thudi et al., 2025), but share our formalism and not our object: none models the performance of a weighted *prediction* ensemble.

The exception, and the closest prior art to this study, is Kwon et al. (2024): the weights of five heterogeneous classifiers as mixture components, a Scheffé canonical polynomial of accuracy or F_1 fitted over a mixture design, reduced by backward elimination and maximized subject to $\sum_i w_i = 1$, on eleven tabular binary datasets. The simplex formulation, the mixture design over classifier weights and the Scheffé surrogate are therefore established prior art for classifiers, and we claim none of them. What that study does not do is what this paper is about: it optimizes one scalar metric at a time, so it has no Pareto front, no anchors and no NBI; it validates the fitted polynomial only in-sample, against no separately sampled reference front; and it carries no cost objective.

2.4 The DoE–RSM–NBI framework and its application to ensemble weights

Normal Boundary Intersection (Das and Dennis, 1998) was introduced to obtain a distributed representation of the Pareto boundary, by solving for each convex weight vector a subproblem that advances from the convex hull of individual minima along a quasi-normal direction. It is widely combined with response-surface metamodels in engineering optimization, and a substantial body of work applies NBI to polynomial surfaces fitted from designed experiments (Brito et al., 2014; Costa et al., 2016; Lopes et al., 2016; Naves et al., 2017; Belinato et al., 2019; de Almeida et al., 2022).

Within that tradition, mixture designs have been used to schedule the NBI weights and to model post-Pareto criteria over the weight simplex. Mendes et al. (2016) and Leal et al. (2022) model portfolio proportions with mixture designs; Bacci et al. (2019) builds a simplex lattice over time-series forecast-combination weights, fits Scheffé models to factor scores of residual accuracy metrics and applies NBI to those models; de Paula et al. (2019) crosses a mixture design over objective weights with genetic-algorithm hyperparameters. Pereira et al. (2025), of which the present first author is a co-author, extends the framework with a mixture-design post-Pareto stage in which Scheffé polynomials of a generalized distance and of the entropy of the weights are themselves optimized by a second NBI. Its anchors, like those of the rest of this literature, are the individual optima of the surrogate surfaces — the configuration we label NBI-A. The first author’s master’s dissertation (Ribeiro, 2026) applies the same framework to hyperparameter optimization.

Most directly, Moreira et al. (2021) determine neural-network ensemble weights by a mixture design, and Rocha et al. (2025) combine a $\{3, 5\}$ simplex-lattice design over the mixing weights of three neural networks, a factor-analytic surrogate of correlated error metrics, NBI over those surrogates, and an entropy-based post-Pareto choice, for short-term load forecasting. That is the present pipeline applied to a regression ensemble, and its stated first contribution is casting ensemble-weight definition as a structured mixture-design problem.

We therefore claim no part of this construction as new. Our contribution begins where this literature stops: none of these studies validates its mixture surrogate on unseen compositions, computes anchors from anything other than the surrogate, revalidates its Pareto candidates on the true objectives, scores them against a separately sampled reference, or replicates over resampled partitions.

2.5 Surrogate-assisted multiobjective optimization

The broader surrogate-assisted literature is well surveyed (Jin, 2011; Tabatabaei et al., 2015; Chugh et al., 2019; Deb et al., 2019), with ParEGO, MOEA/D-EGO and SMS-EGO prominent (Knowles, 2006; Zhang et al., 2010; Ponweiser et al., 2008); it is acutely aware that a surrogate may mislead, and answers primarily with *model management*: deciding which points to evaluate expensively so that the surrogate improves where it matters.

That answer is an *iterative* one, and the distinction bounds everything this paper reports. ParEGO, MOEA/D-EGO, SMS-EGO and K-RVEA evaluate their infill points on the true objectives inside the search loop and refit as they go (Knowles, 2006; Zhang et al., 2010; Ponweiser et al., 2008; Chugh et al., 2018), so a false optimum of the surrogate is corrected by later iterations. The pipeline studied here is the other member of the taxonomy: Tabatabaei et al. (2015) name the *sequential* framework — fit the surrogate once from a fixed design, then optimize it, with no infill — and identify it as the configuration most exposed to surrogate error. That is exactly what the lineage of Section 2.4 does and what we replicate, so our diagnosis is a diagnosis of *one-shot, no-infill* surrogate pipelines. It is not a test of model-managed, infill-based surrogate-assisted optimizers and must not be read as one: by the standards of that literature the exposure of the surrogate-anchored arm is anticipated rather than surprising, and our real-anchor arm is best understood as a single targeted correction — one model-management step

applied to the anchors alone — rather than as adaptive infill.

Extreme-point reliability has been studied, but on the other side of the surrogate divide. Isermann and Steuer (1988) demonstrated computationally that the minima read off a payoff table built from individual optima can differ substantially from the true minimum criterion values over the efficient set; nadir estimation is a recognized problem (Deb et al., 2010); inaccurate estimation of the front range degrades performance across MOEAs (He et al., 2021); and Herrmann-Wicklmayr and Flåßkamp (2026) treat the *choice* of individual minima as a design decision in its own right, replacing extreme minima with non-extreme ones to refine the utopia–nadir hyperbox. That misplaced anchors damage a scalarization is therefore established, and we claim neither the mechanism nor its direction as a discovery. In all of that work the error arises from degeneracy or search difficulty on the *true* objectives; a metamodel is never the culprit. The normalized normal constraint (Messac et al., 2003) was introduced partly to make anchor-based normalization better behaved and is the textbook remedy for the pathology we report; our protocol was frozen with NBI as the front generator, so we do not evaluate it here and record it as the most direct methodological follow-up.

What we did not find is a study in which a CHIM method’s anchors are computed both from a fitted surrogate and from the true objectives on the same problem and the resulting fronts compared — the surrogate *provenance* of the anchor error, and the controlled contrast, rather than the fact that anchors matter. The closest external analogue we located, Abolghasemian et al. (2022), runs a modified NBI on regression metamodels fitted from a designed experiment and validated by fit statistics, without re-evaluating the resulting Pareto solutions on the simulation — precisely the NBI-A pattern this study is built to expose. At the other extreme, Gellerich and Majschak (2023) derive the experimental plan from the NBI subproblems themselves and evaluate each subproblem by physical test, with no metamodel anywhere and anchors that are real experimental optima. A metamodel-free NBI is thus published prior art and the idea of our NBI-C arm is not claimed as new; because that study has no surrogate counterpart on its problem, it makes no comparison. The two papers bracket our NBI-A and NBI-C arms without either of them running the contrast.

2.6 Cost-aware ensemble deployment

Prediction-time cost has been an explicit objective since Turney (2000), priced in three ways: as a fixed *deployed set*, in ensemble pruning and hardware- or energy-aware selection (Zhou et al., 2002; Ji and Li, 2023; Maier et al., 2024; Maier and Purucker, 2026; Gunasekaran et al., 2022; Moreira et al., 2026); as *per-example adaptive execution* through cascades and early exits (Xu et al., 2013; Trapeznikov et al., 2013; Nan and Saligrama, 2017; Chen et al., 2020); and as a *continuous penalty* inside a scalarized training objective (Xu et al., 2012; Zhang and Zhou, 2011). Automated machine-learning systems increasingly expose inference-time constraints during ensemble construction (Erickson et al., 2020; Feurer et al., 2022; Borchert et al., 2022). For post-hoc ensembles over cached tabular predictions — our own setting — Maier et al. (2024) already return a Pareto front of accurate and efficient ensembles with inference time in the objective vector over 83 tabular classification datasets, and Maier and Purucker (2026) extend that line, name the objective *deployment cost* and report memory usage as their most effective cost metric. Inference cost as an objective of a post-hoc weighted ensemble is therefore prior art, and we claim no part of it.

The distinction we need falls between these categories, and that the accounting itself can reorder a comparison is already known: Wang et al. (2022) place two cost accountings side by side — every member executed, versus early exit — and show that which committee looks best changes with the accounting. Their contrast, however, is between ensemble and cascade *architectures*, not between two cost functions applied to one and the same weight vector. Narrowed to that resolution, to the best of our literature search no prior work contrasts the step cost a deployed support pays with the continuous relaxation the optimizers minimize *for one weight vector of*

one *weighted soft-voting ensemble*, and reports that the two disagree about which method wins. Where a related question has been settled it went the other way: for budgeted learning with feature re-use, the support-cost programme is provably solved by its linear relaxation (Nan et al., 2015). Our setting does not inherit that guarantee, and Section 5.7 shows the two definitions disagreeing.

2.7 The gap

Ensemble weights on a simplex, mixture designs over them, and Scheffé surrogates of ensemble performance all exist, and for classifier ensembles specifically they exist together in Kwon et al. (2024); adding NBI on those surrogates exists in Rocha et al. (2025) for a regression ensemble and in Bacci et al. (2019) for forecast combination. Multiobjective optimization of continuous ensemble weights exists. Cost-aware ensemble selection with deployment cost as a Pareto objective exists (Maier et al., 2024; Maier and Purucker, 2026). Metamodel-free NBI on measured objectives exists (Gellerich and Majschak, 2023), and that anchor misplacement degrades a scalarization is established (Isermann and Steuer, 1988; Deb et al., 2010; Herrmann-Wicklmayr and Flaßkamp, 2026). What does not exist, to the best of our literature search, is an evaluation of whether this pipeline can be trusted once it is both multiobjective and applied to classifiers: no external validation of the mixture surrogate on unseen compositions, no computation of the anchors both ways on one problem so that surrogate-derived and real anchors can be contrasted, no metamodel-free arm run alongside the surrogate arms as a control, no revalidation of candidates on the true objectives against an empirical reference with a method-independent core, no replicated partition study of the whole pipeline, and no contrast, for one and the same weight vector, between the cost that is optimized and the cost that is paid.

The gap is therefore not a missing method. It is the absence of the controls that would tell a practitioner when the existing method’s output can be believed — which is what this paper supplies, for the one-shot, no-infill form of the pipeline described above.

3 Methodology

Figure 1 shows the pipeline of this study, marking each element as inherited, adapted for compositional decision variables, or new here.

3.1 The ensemble-weighting problem is a mixture problem

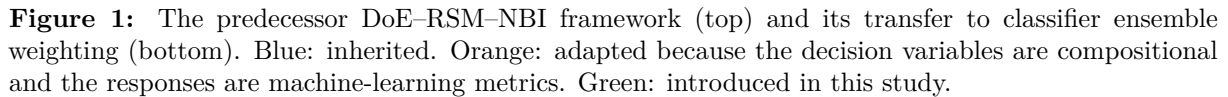
Let M probabilistic base classifiers produce, for a sample x , probability estimates $p_1(x), \dots, p_M(x)$ in $[0, 1]$. A weighted soft-voting ensemble predicts

$$p_{\mathbf{w}}(x) = \sum_{i=1}^M w_i p_i(x) = \mathbf{P}\mathbf{w}, \quad (1)$$

where $\mathbf{P} \in [0, 1]^{n \times M}$ collects the base-model probabilities for n samples. For $p_{\mathbf{w}}(x)$ to remain a probability for every input, the weights must satisfy

$$w_i \geq 0 \quad (i = 1, \dots, M), \quad \sum_{i=1}^M w_i = 1, \quad (2)$$

that is, $\mathbf{w} \in \Delta^{M-1}$, the unit simplex in $\mathbb{R}^M$ and the canonical constraint set of a *mixture experiment*: the components are the classifiers, the vertices are single models used alone, and the centroid is uniform soft voting. The sum-to-one constraint makes an intercept model singular, so the formalism is the mixture design with a Scheffé canonical polynomial, not a factorial design. This correspondence, rather than any performance argument, is why the framework transfers at all.


$$f_1(\mathbf{w}) = -\text{AUC}(y, \mathbf{P}\mathbf{w}) \quad (\text{discrimination, maximized}), \quad (3)$$

$$f_3(\mathbf{w}) = c(\mathbf{w}) \quad (\text{inference cost, minimized}). \quad (5)$$

The log-loss f_2 is convex and smooth in $\mathbf{w}$, whereas the area under the ROC curve is a rank statistic: f_1 is piecewise constant in $\mathbf{w}$.

Let c_i be the measured inference cost of model i (Section 4.2). Deployment cost is a step function of the weight vector: a model receiving *any* non-negligible weight must be loaded and executed; one receiving none costs nothing. This gives the *support cost*

$$c_{\text{sup}}(\mathbf{w}) = \sum_{i=1}^M c_i \mathbf{1}[w_i > \varepsilon], \quad \varepsilon = 10^{-3}, \quad (6)$$

$$c_w(\mathbf{w}) = \sum_{i=1}^M w_i c_i, \quad (7)$$

9

3.3 Mixture design and Scheffé surrogates

Design. With $M = 5$ the design is a fixed 66-run mixture design: the union of the $\{5, 3\}$ and $\{5, 2\}$ simplex lattices, the overall centroid, the 5 axial points, the 5 quaternary centroids and the 10 midpoints between the overall centroid and the ternary centroids. It therefore places design points on the 5 pure vertices, all 10 binary edges, the ternary and quaternary faces and uniform voting, supporting the surrogate on every face a sparse ensemble could occupy.

Models. For a response y the Scheffé canonical polynomials of order 1, 2 and the special cubic are

$$\text{linear: } \hat{y} = \sum_i \beta_i w_i, \quad (8)$$

$$\text{quadratic: } \hat{y} = \sum_i \beta_i w_i + \sum_{i < j} \beta_{ij} w_i w_j, \quad (9)$$

$$\text{special cubic: } \hat{y} = \sum_i \beta_i w_i + \sum_{i < j} \beta_{ij} w_i w_j + \sum_{i < j < k} \beta_{ijk} w_i w_j w_k, \quad (10)$$

fitted by ordinary least squares without an intercept and without pure quadratic terms, which $\sum_i w_i = 1$ makes redundant. No term elimination is performed: the canonical form is kept intact so that β_i is the predicted response at vertex i .

External validation and the reliability gate. Each replication also evaluates 100 compositions drawn from Dirichlet(1) (60) and Dirichlet(0.5·1) (40) that are not in the design and never used for fitting, giving an external R_{ext}^2 and Spearman ρ_{ext} . *Order selection* uses a parsimony rule — the lowest order whose external RMSE is within 10% of the best — so it is driven by out-of-design accuracy rather than in-sample fit. The *same* 100 compositions serve both steps: the parsimony rule picks the order on them and the gate then scores the selected order on them. The gate is therefore not independent of order selection — the order it judges was chosen to do well on exactly these points — and its pass rate should be read as an optimistic estimate of out-of-design adequacy. We report alongside it the external RMSE of the selected surface in the response’s own units and the observed range of that response over the same 100 points (Table 2), because on a nearly flat surface R_{ext}^2 is ill-posed — its denominator is the variance of an almost constant response — so the RMSE read against the range is the more informative adequacy measure. *Admissibility* uses a *reliability gate* whose thresholds were fixed and committed to version control before the replicated benchmark was run, and never adjusted afterwards: a surface is reliable only if

$$R_{\text{ext}}^2 \geq 0.5 \quad \text{and} \quad \rho_{\text{ext}} \geq 0.9. \quad (11)$$

We describe this as pre-specification rather than preregistration: there is no external registry entry, and the evidence is the version-controlled commit history. The gate is a hypothesis under test, not a filter applied to the results: surrogate-based optimization is run and reported *regardless* of the gate outcome, so that its predictive value can be measured (Section 5.1).

3.4 Normal Boundary Intersection on the simplex

NBI (Das and Dennis, 1998) was introduced to obtain a distributed representation of the Pareto boundary (Section 2.4). Whether that distribution survives the map into revalidated objective space is one of the questions this paper tests, and Section 5.4 reports that in our setting it does not. For m objectives $F = (F_1, \dots, F_m)$ let $\mathbf{w}^{*(k)}$ minimize F_k ; the payoff matrix Φ has columns $F(\mathbf{w}^{*(k)})$, its diagonal is the utopia point F^u , and objectives are normalized by the utopia and pseudo-nadir points extracted from Φ . Writing $\bar{\Phi}$ for the normalized payoff matrix and $\hat{n} = -\bar{\Phi}\mathbf{1}$

for the quasi-normal direction, each β on the unit simplex of *scalarization* weights defines the subproblem

$$\max_{\mathbf{w}, t} \quad t \quad \text{s.t.} \quad \bar{\Phi}\beta + t\hat{n} = \bar{F}(\mathbf{w}), \quad \mathbf{w} \in \Delta^{M-1}. \quad (12)$$

The simplex constraint is eliminated by optimizing over $M - 1$ free variables with $w_M = 1 - \sum_{i < M} w_i$, with a deterministic feasibility-recovery step for iterates that leave the simplex: the free variables are clipped to $[0, 1]$ and, if they still sum above one, renormalized onto the face $w_M = 0$. This is a documented repair, not a Euclidean projection onto the simplex, and a candidate recovered this way is flagged infeasible and excluded from the certified set. Without it, subproblems near a vertex terminate outside the feasible region. For vertex values of β the solution is the corresponding anchor, returned directly rather than re-derived. All variants solve 66 subproblems per replication on the same β lattice, so the arms are compared at identical points of the CHIM.

The three variants. The variants are defined by two methodological factors — the source of the objectives and the source of the anchors:

Variant	Objectives in Equation (12)	Anchors / payoff matrix	Real evaluations, standalone
NBI-A	Scheffé surfaces	optima of the Scheffé surfaces	1.7×10^2
NBI-B	Scheffé surfaces	real out-of-fold optima	$\approx 4.1 \times 10^4$
NBI-C	real out-of-fold objectives	real out-of-fold optima	$\approx 4.6 \times 10^5$

NBI-A is the lineage construction: everything, including the payoff matrix, comes from the metamodel. NBI-B changes only the anchors, which become the real optima of Section 3.5; NBI-C removes the metamodel entirely. The last column charges every variant for each real objective evaluation it needs *standalone*: the 66 design points and the 100 gate compositions for NBI-A; those plus the real anchors for NBI-B, dominated by the direct ROC-AUC search over 4×10^4 sampled compositions (Section 3.5); and the real anchors plus its own solve for NBI-C. Real anchors are not free. NBI-B consumes about 4×10^4 real evaluations that NBI-A does not — roughly a tenth of NBI-C’s entire budget — and its standalone wall clock is 1.2–12.7 times NBI-A’s, which is why Table 5 prices each arm for the stages it depends on rather than for its solver stage alone. Because f_1 is piecewise constant, NBI-C uses finite-difference steps of 10^{-3} , multistart on the simplex, and accepts an iterate whose equality residual is below 10^{-3} even without certified optimality, whereas NBI-A and NBI-B optimize smooth polynomials and require full SLSQP certification. This asymmetry follows from the objective, not from tuning, and it bounds the causal reading. A versus B is not a single-factor contrast: replacing the anchors simultaneously moves the utopia point, rescales both normalization axes through the payoff matrix, reorients the CHIM and every quasi-normal direction, and—because the three vertex values of β return their anchor directly—substitutes three returned candidates. Section 5.3 separates the last of these from the rest. B versus C tests removal of *surrogate error in the interior* once real anchors are supplied, but is not a perfectly identical numerical experiment, because changing the objective from a smooth polynomial to a piecewise-constant rank statistic necessarily changes solver semantics. Solver outcomes are therefore reported separately for each arm.

3.5 References, comparators and revalidation

Single-objective references. Each replication computes the best single model, uniform voting, logistic stacking, the SLSQP log-loss optimum (convex, hence global), a derivative-free ROC-AUC optimum over 4×10^4 sampled compositions with local refinement, and the optima of the two Scheffé surfaces. The log-loss and AUC optima double as the real anchors of NBI-B and NBI-C. SLSQP on the convex log-loss remains the correct tool for the single-objective problem; NBI is not proposed as a replacement for it.

Comparators. Two cheap alternatives are run on every replication: random weighted scalarization, which draws 66 scalarization weight vectors from `Dirichlet(1)` and minimizes the corresponding weighted sum of the *same* Scheffé surfaces with the same real anchors used by NBI-B; and a 66-point random `Dirichlet(1)` sample of the weight simplex. The second is matched to the NBI sets only in the *number of returned candidates* and is not compute-matched; it functions as a floor check on the geometry of the reference front, not as an efficiency baseline.

Revalidation. Every candidate from every method — the three NBI arms, both comparators, the 66 design runs and the references — is re-evaluated on the exact out-of-fold objectives before any indicator is computed, with non-dominated filtering applied to those revalidated values. No result in this paper is read off a surrogate.

3.6 Empirical Pareto reference and quality indicators

The true Pareto set of Equation (1) over Δ^{M-1} is unknown, so each replication constructs an *empirical Pareto reference*: at least 10^5 compositions from `Dirichlet(1)` and `Dirichlet(0.3 · 1)` in equal parts, augmented with the simplex lattice, all vertices and edges, and a 40-cap ε -constraint sweep, evaluated on the real objectives and filtered to the non-dominated set. An independent sample of 2×10^4 points checks convergence: if it displaces more than 5% of the front, the reference is enlarged and rechecked, up to three rounds. This sampled core contains no output of the five methods compared in this study — NBI-A, NBI-B, NBI-C, random weighted scalarization and NSGA-II. It is not, however, free of optimization output: it carries the 40 ε -constraint solutions and the seven single-objective references of Section 3.5, among them the real ROC-AUC and log-loss optima that NBI-B and NBI-C receive as anchors and the two Scheffé-surface optima that NBI-A receives. On Santander the ε -constraint sweep supplies a median of 38 of the roughly 135 core-front points and the single-objective references a median of 4 (`tables/reference_diagnostics.csv`). The final reference used for scoring is the non-dominated union of that core with every candidate set of the replication, making it at least as good as any method scored against it.

We call this an empirical reference, never a true Pareto front: it is a sample-based upper bound, and because candidate sets contribute to it, a method contributing many points is partly graded against itself.

Approximation sets are scored on min–max normalized objectives by generational distance, inverted generational distance, IGD^+ (Ishibuchi et al., 2015), Schott spacing, the exact hypervolume with reference point 1.1 on each axis, expressed as a ratio to the reference hypervolume, coverage, and the joint non-dominated fraction.

3.7 The normalization box

Every indicator is computed on objectives mapped into $[0, 1]$ by one box, and the box is not a fixed property of the problem. Let $\mathcal{F}$ be the empirical reference front of the replication — the non-dominated union of the sampled core with every candidate set — under one of the two cost definitions. The box is the componentwise minimum and maximum of $\mathcal{F}$:

$$\ell_k = \min_{F \in \mathcal{F}} F_k, \quad u_k = \max_{F \in \mathcal{F}} F_k, \quad \tilde{F}_k = \frac{F_k - \ell_k}{u_k - \ell_k}, \quad k = 1, 2, 3. \quad (13)$$

Three consequences follow, and each constrains how the numbers in Section 5 may be read.

(i) *Within a replication the box is common to every arm.* All candidate sets of a replication are scored under the same ℓ and u , so the paired contrast between two arms is a difference of commensurable quantities and is unaffected by the choice of box.

(ii) *The box changes when a method is added to the reference,* because a new candidate set can extend $\mathcal{F}$. Adding the NSGA-II arm moves every indicator level slightly: on the identical

Table 1: Datasets. Rows used, number of raw features, positive-class prevalence and training/holdout sizes of each outer partition (stratified 80/20). Kaggle test labels are never used; all selection uses 5-fold out-of-fold predictions within the training part. [†]200,000-row stratified subsample of the 595,212 available rows, drawn once with a recorded seed.

Dataset	Source	Rows	Features	Prevalence	Train / holdout rows
Santander	Santander Customer Transaction (Kaggle 2019)	200,000	200	0.100	160,000 / 40,000
BNP Paribas	BNP Paribas Cardif Claims (Kaggle 2016)	114,321	131	0.761	91,456 / 22,865
Porto Seguro	Porto Seguro Safe Driver (Kaggle 2017) [†]	200,000	57	0.036	160,000 / 40,000
UCI credit	Default of credit card clients (UCI 350)	30,000	23	0.221	24,000 / 6,000

candidate sets, NBI-C’s median hypervolume ratio is 0.989, 0.983, 0.982 and 0.976 in Section 5.4, computed without NSGA-II in the reference, and 0.990, 0.981, 0.981 and 0.977 in Table 6, computed with it. Absolute indicator levels are therefore only interpretable together with the set of methods that built the reference.

(iii) *The box differs between replications*, since each replication has its own reference front. Absolute levels are comparable within a replication but not across partitions, and a mean of ΔHV over 30 partitions is a mean over 30 different scales. Paired within-replication differences are unaffected, which is why medians, win counts and per-partition distributions are reported alongside means; Section 5.7 withdraws one earlier between-partition reading that this rescaling, not the optimizers, produced.

4 Experimental protocol

The protocol was fixed before the replicated study and was not changed after any result was observed; the extension from 10 to 30 replications reused it without modification, and Section 5.9 reports what that extension changed.

4.1 Datasets

Four public binary tabular classification problems span prevalence, dimensionality and feature type (Table 1): Santander (high-dimensional, anonymized, numeric, 10% positives); BNP Paribas (numeric plus high-cardinality categorical features, 76% positive); Porto Seguro (strongly imbalanced at 3.6% positives, subsampled to 200,000 rows by a single stratified draw with a recorded seed to bound runtime); and UCI credit default (small and low-dimensional). For the three Kaggle competition datasets the competition test labels are not public and were never used: all training, weighting, thresholding and model selection use only the published training file, from which every performance figure in this paper is derived.

4.2 Base models and leakage control

The ensemble contains $M = 5$ deliberately heterogeneous learners with fixed hyperparameters and no hyperparameter search: ℓ_2 -regularized logistic regression; Gaussian naive Bayes; k -nearest neighbours with $k = 100$; a random forest with 200 trees and a minimum of 20 samples per leaf; and gradient-boosted trees (XGBoost, 400 rounds, depth 6, learning rate 0.1, subsample and column subsample 0.8). The zoo spans the accuracy–cost–calibration space rather than forming a competitive stack: it deliberately includes a member strong in ranking but poorly calibrated (Gaussian naive Bayes) and one accurate but two orders of magnitude more expensive at inference time (k -nearest neighbours). Preprocessing — imputation, standardization and encoding, detailed per model in the supplement — is fitted inside each training fold and never on validation data.

Each replication splits the dataset by a stratified 80/20 partition into a training part and an untouched holdout. Within the training part, stratified 5-fold cross-validation produces out-

of-fold probability estimates $\mathbf{P}$ for all five models; these are the only quantities the design, the surfaces, the optimizers and the selection rules ever see. Each model is then refitted once on the whole training part to produce holdout probabilities $\mathbf{Q}$, used exclusively for the confirmation analysis of Section 5.8. Because Equation (1) is linear in $\mathbf{w}$, evaluating any weight vector costs one matrix–vector product, which is what makes an empirical reference of 10^5 points and a metamodel-free NBI arm affordable.

Inference cost c_i is measured, not assumed: per model and replication, the median of five timed `predict_proba` calls on a fixed 10,000-row batch, including preprocessing, in milliseconds per 1,000 rows.

4.3 Replication, seeds and the statistical unit

The entire pipeline, from the outer partition through scoring, is executed $R = 30$ times per dataset, giving 120 replications, with the outer partition, the inner folds and every stochastic component seeded by $20260904 + r$ for replication r . All seeds are persisted in the manifest, and the checkpointed runner of Section 8 recomputes any stage exactly.

The statistical unit is **one outer partition of one dataset**. Because all methods are evaluated on the same partition, every comparison is *paired*. Two consequences constrain the analysis throughout.

First, the 30 partitions of a dataset overlap heavily: two independent stratified 80/20 draws have an expected intersection of $0.8 \times 0.8 = 64\%$ of the dataset, which is 80% of each training part, a Jaccard index of two-thirds. Any two training parts therefore share most of their rows. The replication distribution therefore measures *sensitivity to data partitioning*, not sampling from a population, and the effective information is far below 30 independent runs. Second, and consequently, the dataset is the unit of generalization: results are summarized per dataset and the four dataset-level results are then compared. Nothing is pooled across the 4×30 replications, and no statement of the form “ $N = 120$ independent experiments” is made anywhere in this paper.

4.4 Statistical analysis

The primary comparisons were fixed before the replicated benchmark and committed to version control: NBI-B versus NBI-A, NBI-C versus NBI-B, NBI-C versus random scalarization, and NBI-C versus the random Dirichlet sample. The primary endpoints are IGD^+ and the hypervolume ratio, computed on the revalidated *out-of-fold objectives of the 80% training part* — the only quantities the design, the surfaces, the optimizers and the selection rules ever see (Section 4.2) — against the empirical reference of the same replication, under the weighted cost that the optimizers actually minimized; the support-cost repetition is a pre-specified sensitivity analysis. The untouched 20% holdout enters no primary endpoint: it is used only for the confirmation analysis of Section 5.8. All other indicators are descriptive.

Paired differences are signed so that $\Delta > 0$ always favours the second-named method: $\Delta\text{HV} = \text{HV}_{\text{new}} - \text{HV}_{\text{ref}}$ and $\Delta\text{IGD}^+ = \text{IGD}_{\text{ref}}^+ - \text{IGD}_{\text{new}}^+$. For each dataset $\times$ comparison $\times$ endpoint we report the mean and median with percentile-bootstrap 95% intervals (10,000 resamples, fixed seed), the quartiles and range, the win/tie/loss counts with a tie tolerance of 10^{-4} , the win fraction with a Wilson 95% interval, and the matched-pairs rank-biserial correlation as a distribution-free effect size.

An ordinary paired t -test on resampled partitions is anticonservative, so the primary test is the Nadeau–Bengio corrected resampled t -test (Nadeau and Bengio, 2003): with $J = 30$ paired differences d_j ,

$$t = \frac{\bar{d}}{\sqrt{\text{var}(d) \left(\frac{1}{J} + \rho \right)}}, \quad \rho = \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{0.20}{0.80} = 0.25, \quad (14)$$

Table 2: Surrogate reliability over 30 partitions, all quantities measured on the *same* 100 unseen compositions: the parsimony rule selects the Scheffé order on them (lowest order within 10% of the best external RMSE) and the reliability gate ($R^2_{\text{ext}} \geq 0.5$ and Spearman $\rho \geq 0.9$) is then scored on them, so the gate is not independent of order selection. Orders are abbreviated lin./quad./sp. cubic. Pass counts carry Wilson 95% intervals and the median external R^2 its bootstrap interval. The last two columns give the median external RMSE of the selected surface in the response’s own units and the median observed range of that response over the same 100 points: on a flat surface R^2_{ext} is ill-posed, so the RMSE read against the range is the more informative adequacy measure. Sources: `statistics/reliability_gate_r30.csv`, `tables/scheffe_orders.csv`.

Dataset	Response	Selected order (count)	Pass	Wilson 95%	Median R^2_{ext} [95% CI]	Median ρ	RMSE _{ext}	Range
Santander	ROC-AUC	quad. 26, lin. 4	0/30	[0.00, 0.11]	-0.333 [-0.559, -0.199]	0.830	0.0087	0.040
Santander	log-loss	quad. 29, sp. cubic 1	30/30	[0.89, 1.00]	0.971 [0.967, 0.973]	0.994	0.0028	0.081
BNP Paribas	ROC-AUC	lin. 22, quad. 7, sp. cubic 1	5/30	[0.07, 0.34]	0.118 [-0.034, 0.244]	0.908	0.0096	0.053
BNP Paribas	log-loss	lin. 21, quad. 7, sp. cubic 2	3/30	[0.03, 0.26]	-0.180 [-0.246, 0.042]	0.955	0.1214	0.694
Porto Seguro	ROC-AUC	quad. 26, sp. cubic 2, lin. 2	16/30	[0.36, 0.70]	0.519 [0.482, 0.581]	0.924	0.0037	0.030
Porto Seguro	log-loss	lin. 20, quad. 8, sp. cubic 2	12/30	[0.25, 0.58]	0.506 [0.492, 0.580]	0.995	0.0582	0.448
UCI credit	ROC-AUC	quad. 20, sp. cubic 10	30/30	[0.89, 1.00]	0.992 [0.989, 0.992]	0.995	0.0015	0.077
UCI credit	log-loss	quad. 27, sp. cubic 3	30/30	[0.89, 1.00]	0.993 [0.991, 0.993]	0.993	0.0066	0.427

on $J - 1$ degrees of freedom. The correction inflates the variance by a factor of $(1/30 + 0.25)/(1/30) \approx 8.5$, multiplying standard errors by about 2.9 relative to the naive test. One caveat attaches to ρ and we state it plainly: the heuristic $\rho = n_{\text{test}}/n_{\text{train}}$ was derived for a statistic measured on the held-out portion of each resample, whereas both primary endpoints here are measured on the training part, whose rows two replications share to the extent computed above. The value $\rho = 0.25$ is therefore a lower bound on the dependence between replications and the corrected test is, if anything, still anticonservative; every significance statement below should be read as holding *at* $\rho = 0.25$, with the effect consistency that does not depend on ρ reported alongside it. Within each dataset the eight primary tests form one family and receive a Holm correction; no correction is applied across datasets, because datasets are not pooled. A Wilcoxon signed-rank test is reported alongside as a distribution-free check, but it is not corrected for overlap and cannot rescue a non-significant corrected result.

Where a paired-difference distribution is visibly bimodal or heavy-tailed, the mean-based test is uninformative by construction and we say so, reporting the median with its bootstrap interval, the win fraction with its Wilson interval and the rank-biserial correlation as the evidence instead. Throughout, effect consistency takes precedence over p -values. Frequencies over replications are reported with Wilson and Jeffreys 95% intervals.

5 Results

Results are organized by research question, with per-claim provenance in the supplementary claims map. Indicators use the weighted cost of Equation (7) unless stated otherwise.

One fact frames the rest: weighting helps, but modestly. The log-loss optimum improves out-of-fold ROC-AUC over the best single model by 0.0037, 0.0056, 0.0060 and 0.0007 — one to two orders of magnitude more than the differences between the multiobjective methods studied here.

5.1 RQ1: Accuracy of the Scheffé surfaces

Surrogate adequacy depends on dataset and response alike (Table 2, Figure 2). The ROC-AUC surface passes the gate in 0/30 partitions on Santander, 5/30 on BNP Paribas, 16/30 on Porto Seguro and 30/30 on UCI credit; the log-loss surface passes in 30/30, 3/30, 12/30 and 30/30. Santander is sharpest: on the same partition its log-loss surface is essentially exact (median $R^2_{\text{ext}} = 0.971$) while its ROC-AUC surface has *negative* external R^2_{ext} (median -0.333) — worse than predicting the mean — even though its external rank correlation is a respectable 0.830.

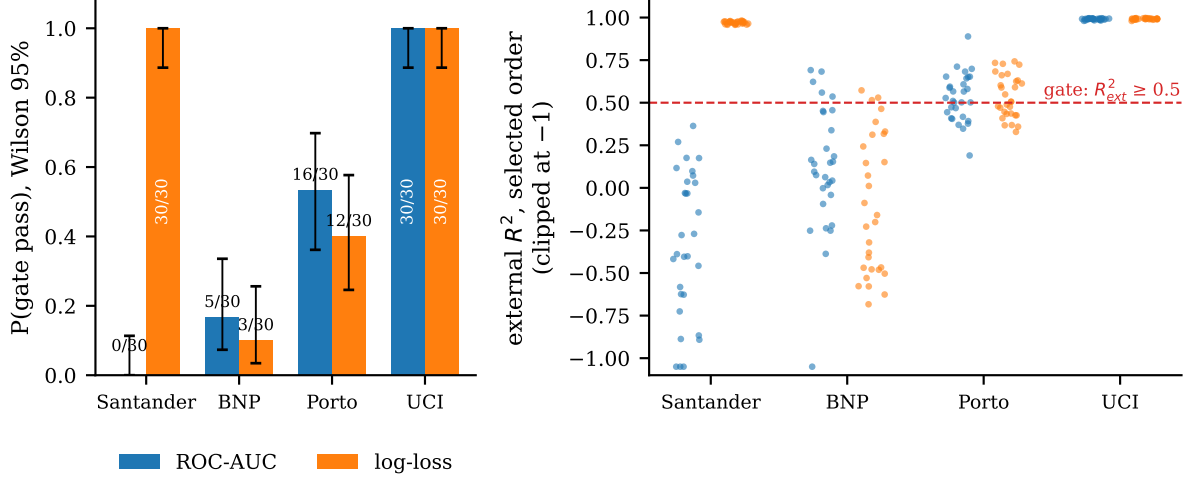


Figure 2: Left: reliability-gate pass frequency per dataset and response over 30 partitions, with Wilson 95% intervals. Right: the external R^2_{ext} of the selected order per partition; the dashed line is the gate threshold. Within Santander the two responses fall on opposite sides of the gate.

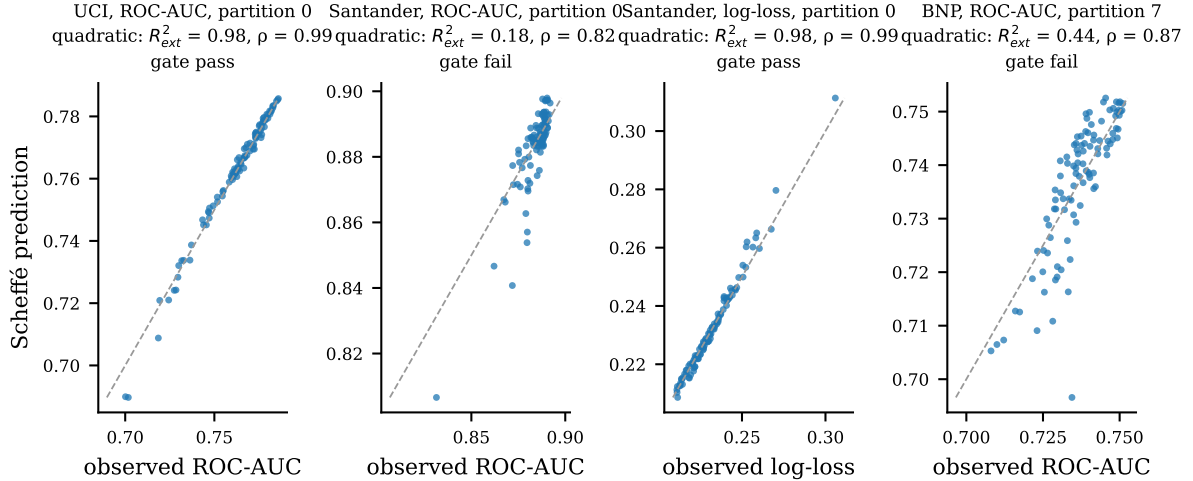


Figure 3: Predicted versus observed response on the 100 unseen Dirichlet compositions of one partition, for four dataset–response pairs spanning the regimes of Table 2. The dashed line is the identity.

Porto Seguro sits on the threshold for both responses and 30 partitions do not resolve it either way — a property of the problem, not a sample-size deficiency.

The failures are misspecification, not extrapolation: the excess of predicted over observed range on the validation points is approximately zero everywhere, so the surrogate is not predicting outside its fitted region. As a control, the Brier score — genuinely quadratic in $\mathbf{w}$ — is reproduced with $R^2_{\text{ext}} = 1.000$ in all 120 replications, so design and fitting are sound and the failure is specific to the response (Figure 3).

5.2 RQ2: Surface stability and what the coefficients mean

The surfaces are stable across resampling: the four largest interaction coefficients keep their sign in all eight dataset–response cells, the largest-magnitude term is the same at $R = 10$ and $R = 30$ in all eight and is the per-partition largest in 30/30 partitions in seven of them (on Porto Seguro’s ROC-AUC surface two k -nearest-neighbour pairs alternate, 19/30 against 11/30), and between-partition coefficients of variation for the leading terms lie between 0.011 and 0.17.

Table 3: The classical synergism criterion applied correctly, and what the real objectives say. On the binary edge i - j of a quadratic Scheffé model the fitted surface has an interior optimum exceeding both pure components if and only if $\hat{\beta}_{ij} > |\hat{\beta}_i - \hat{\beta}_j|$. Columns 3–5 evaluate that criterion for the largest- $\hat{\beta}$ pair of each dataset; column 6 gives the partitions in which the *real* out-of-fold 50/50 blend of the same pair beats its better member. The last three columns compare criterion and reality across all ten pairs: agreements, cases where the surface predicts a superior blend that does not exist, and the reverse. The error is systematic and one-directional. Source: `tables/edge_condition_summary.csv`.

Dataset	Pair	Largest- $\hat{\beta}$ pair, fitted surface			Real blend	All 10 pairs		
		$\hat{\beta}_{ij}$	$ \hat{\beta}_i - \hat{\beta}_j $	criterion met	beats better member	agree	surface over-predicts	under-predicts
Santander	GNB-kNN	0.243	0.130	30/30	0/30	6/10	4	0
BNP Paribas	GNB-XGB	0.148	0.110	30/30	0/30	4/10	6	0
Porto Seguro	GNB-kNN	0.071	0.025	30/30	30/30	7/10	3	0
UCI credit	GNB-XGB	0.130	0.100	30/30	0/30	3/10	7	0

Interpreting those coefficients requires the classical criterion (Scheffé, 1958; Cornell, 2002) stated correctly. The edge profile $\hat{y}(t) = \beta_j + (\beta_i - \beta_j)t + \beta_{ij}t(1 - t)$ implied by Equation (9) has an interior optimum above both pure components if and only if

$$\beta_{ij} > |\beta_i - \beta_j| \quad (15)$$

(derivation in supplementary Section S3), which already guards against the naive reading: a large β_{ij} on an edge joining two very unequal vertices need not beat the better one. Our finding is not that this criterion is unknown, but that *the fitted surfaces pass it and are still wrong*.

For the largest- β pair of every dataset Equation (15) holds in all thirty partitions (Table 3), yet on the real out-of-fold objectives the predicted superior blend materializes on one dataset only. The real 50/50 blend is *worse* than its better member in 30/30 partitions on Santander (-0.0050 ROC-AUC), BNP Paribas (-0.0180) and UCI credit (-0.0201), and beats both members only on Porto Seguro ($+0.0098$, 30/30); the untouched holdout repeats this (0/30, 0/30, 29/30, 0/30), and optimal rather than 50/50 weighting does not rescue it (supplementary Section S3). Across all ten pairs criterion and reality agree on 6, 4, 7 and 3; all 20 disagreements are the surface predicting a blend the real objectives do not deliver, and none is a real one missed.

The mechanism is visible in Figure 4: the largest interaction attaches to the *weakest* vertex — k -nearest neighbours on Santander and Porto Seguro, Gaussian naive Bayes on BNP Paribas and UCI credit — and tracks the pair’s vertex-quality gap (0.163, 0.145, 0.022 and 0.107). Consistently, the pairs whose real blend does beat its better member in at least 80% of partitions are different, smaller- β pairs on every dataset except Porto Seguro (supplementary Section S3).

The criterion is correct and should be applied through Equation (15) rather than by reading the sign of β_{ij} , but on these ensembles it inherits the misspecification of the surface it is computed from and must be confirmed on the real objective — one matrix product per pair — before a pair is called complementary. Log-loss interactions are uniformly negative and uninformative, since log-loss is convex in $p_{\mathbf{w}}$ and Jensen’s inequality forces blending to look favourable; only the ROC-AUC signs are interpretable, and those are the ones the blend test contradicts.

5.3 RQ3 and RQ4: Surrogate versus real anchors

Replacing the surrogate-derived anchors with anchors computed from the real single-objective optima, changing nothing else, improves both primary endpoints in all 30 partitions on Santander (median $\Delta\text{HV} = +0.196$, median $\Delta\text{IGD}^+ = +0.125$; Holm-corrected $p = 0.004$ and 0.007) and in all 30 on BNP Paribas (medians $+0.058$ and $+0.0146$). On the other two the improvement is not unanimous and the endpoints differ: Porto Seguro 23/30 partitions on IGD^+ (median $+0.0296$) and 24/30 on hypervolume (median $+0.076$), UCI credit 24/30 on IGD^+ (median $+0.0499$) and 24 wins with one tie and five losses on hypervolume (median $+0.092$). Bootstrap intervals for every median are in Table 4 (Figure 5, Figure 6).

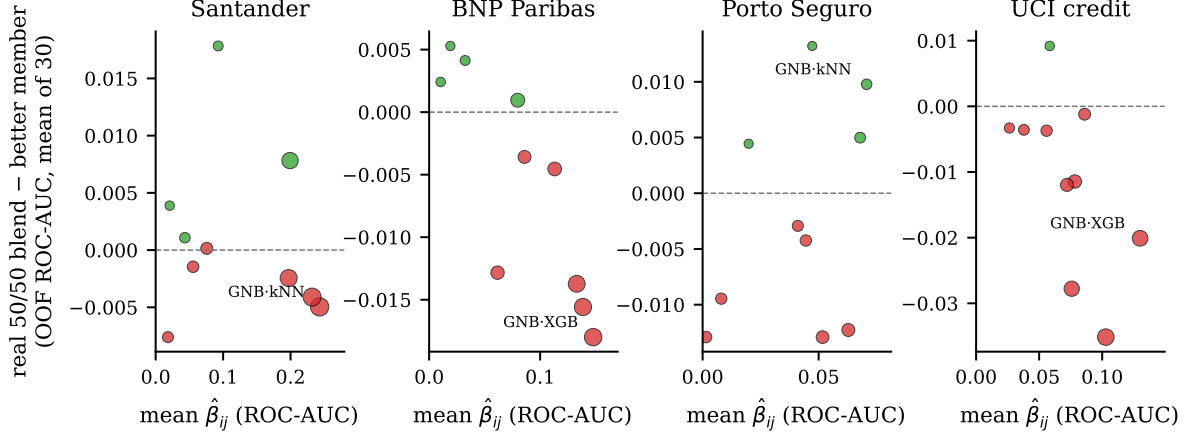


Figure 4: Fitted interaction coefficient against the real gain of the corresponding 50/50 blend over its better member, one point per classifier pair, averaged over 30 partitions; marker size is the vertex-quality gap. Green markers denote pairs whose blend wins in at least 80% of partitions; the largest- β pair is labelled.

Table 4: Primary paired comparisons under the weighted cost, 30 partitions per dataset. $\Delta > 0$ favours the second-named set ($\Delta\text{IGD}^+ = \text{IGD}_{\text{ref}}^+ - \text{IGD}_{\text{new}}^+$, $\Delta\text{HV} = \text{HV}_{\text{new}} - \text{HV}_{\text{ref}}$). Median with its percentile-bootstrap 95% interval, wins/ties/losses, and the Holm-corrected Nadeau–Bengio significance band ($\rho = 0.25$, family of eight tests per dataset; *n.s.* means the corrected test does not reach 0.05, which for a bimodal or heavy-tailed difference is expected and is why the median and the win count are reported). Means, rank-biserial effect sizes, exact p -values, the two floor-check comparisons against random scalarization and random Dirichlet search, and the full support-cost repetition are in supplementary Tables S1 and S5. Source: `statistics/paired_primary_effects.csv`, `paired_primary_tests.csv`.

Dataset	Comparison	ΔIGD^+			Δ hypervolume ratio		
		median [95% CI]	W/T/L	Holm p	median [95% CI]	W/T/L	Holm p
Santander	NBI-B vs NBI-A	+0.125 [+0.086, +0.145]	30/0/0	<0.01	+0.196 [+0.143, +0.216]	30/0/0	<0.01
	NBI-C vs NBI-B	+0.0017 [+0.0011, +0.0020]	28/0/2	n.s.	+0.0084 [+0.0072, +0.0093]	28/0/2	<0.01
BNP Paribas	NBI-B vs NBI-A	+0.0146 [+0.0136, +0.0172]	30/0/0	n.s.	+0.058 [+0.051, +0.070]	30/0/0	n.s.
	NBI-C vs NBI-B	-0.0028 [-0.0044, -0.0015]	6/0/24	n.s.	+0.0106 [+0.0087, +0.0128]	24/0/6	n.s.
Porto Seguro	NBI-B vs NBI-A	+0.0296 [+0.0193, +0.0730]	23/0/7	n.s.	+0.076 [+0.046, +0.157]	24/0/6	n.s.
	NBI-C vs NBI-B	+0.0244 [+0.0168, +0.0408]	28/0/2	n.s.	+0.074 [+0.039, +0.100]	28/0/2	n.s.
UCI credit	NBI-B vs NBI-A	+0.0499 [+0.0327, +0.0606]	24/0/6	n.s.	+0.092 [+0.039, +0.100]	24/1/5	n.s.
	NBI-C vs NBI-B	+0.145 [+0.126, +0.182]	29/0/1	<0.01	+0.275 [+0.219, +0.329]	30/0/0	<0.01

Two readings the win counts invite are not supported by the magnitudes. First, a unanimous count is not a large effect: Santander and BNP Paribas are both 30/30, yet the median hypervolume gain is +0.196 on the first and +0.058 on the second. Second, the benefit does not track surrogate reliability. Ranked by median hypervolume gain the datasets are Santander +0.196 (ROC-AUC gate 0/30), UCI credit +0.092 (30/30), Porto Seguro +0.076 (16/30) and BNP Paribas +0.058 (5/30): the dataset whose ROC-AUC surface never passes the gate shows the largest effect and the dataset whose surfaces always pass shows the second largest. Real anchors helped on all four datasets, including the one whose surfaces are excellent; what is confined to the unreliable end is the collapse mode described below, not the typical magnitude of the benefit.

Table 4 scores every set against the augmented reference of its own replication (Section 3.6). The sample-core repetition of all four primary comparisons is in the supplement (supplementary Section S5, computed from `nsga2/nsga2_pareto_quality.csv`, which scores every candidate set under both reference definitions). It moves the magnitudes without changing any direction here — the Santander hypervolume median falls from +0.196 to +0.145 and the BNP Paribas

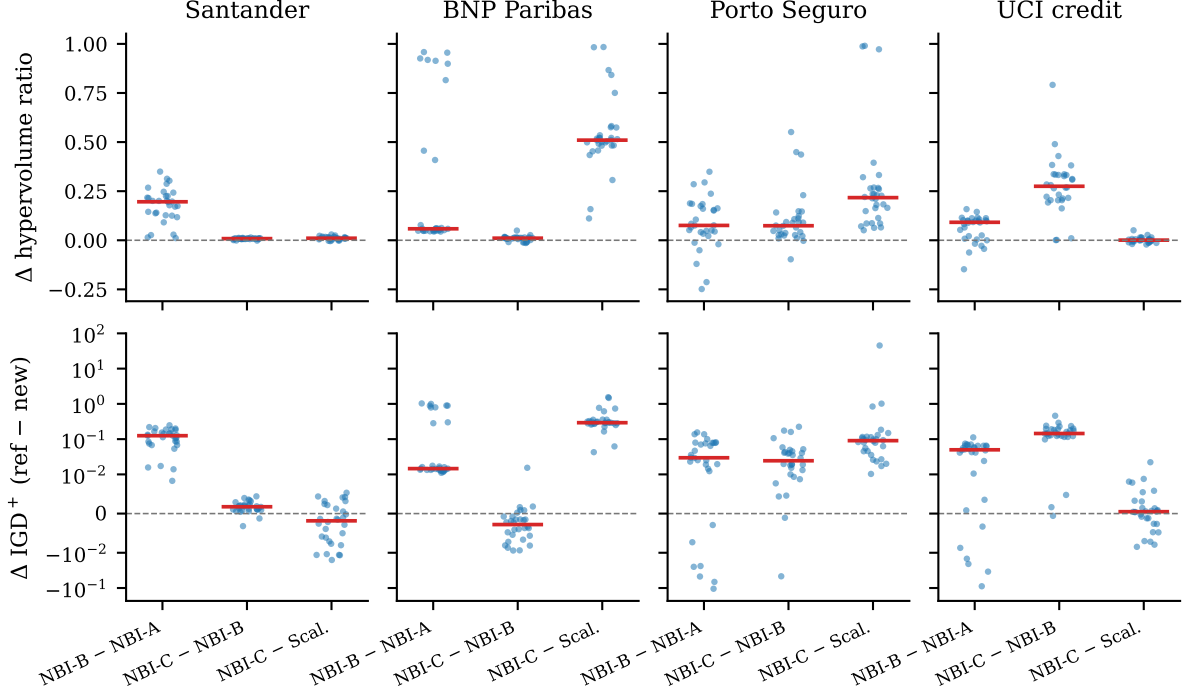


Figure 5: Paired differences on the two primary endpoints, one point per outer partition, red bar at the median; the IGD^+ axis is symmetric-logarithmic. The BNP Paribas NBI-B – NBI-A distribution is visibly bimodal: small positive differences plus a tail of seven partitions where NBI-A collapses.

median rises from $+0.058$ to $+0.070$ — and leaves the ordering above inverted in the same way ($+0.145$, $+0.089$, $+0.078$ and $+0.070$).

Three qualifications bound this. First, corrected significance is the exception here, not the rule. Of the sixteen primary tests that carry the two NBI comparisons — two comparisons $\times$ two endpoints $\times$ four datasets, weighted cost — five reach Holm-corrected $p < 0.05$: NBI-B versus NBI-A on Santander (IGD^+ 0.007, hypervolume 0.004), and NBI-C versus NBI-B on Santander hypervolume (0.003) and on UCI credit (IGD^+ 0.010, hypervolume 0.010). The other eleven do not — every test on BNP Paribas and Porto Seguro, both UCI credit anchor tests, and Santander IGD^+ for NBI-C versus NBI-B. On BNP Paribas the effect is unanimous in sign yet the corrected p is 0.66, because the differences are bimodal — the mean $+0.279$ is over four times the median $+0.058$ — so by the rule fixed in advance we report the median with its bootstrap interval $[+0.051, +0.070]$, the win fraction with its Wilson interval ($[0.89, 1.00]$ for a 30/30 fraction) and the rank-biserial correlation (1.00 on the two unanimous datasets, 0.65–0.75 on the other two). Those three quantities describe how *consistent* a difference is across the 30 partitions of a dataset; none of them corrects for the partition overlap of Section 4.4, which is what the corrected test exists to handle, so they do not establish significance and are not offered in its place. Where the corrected test does not reach 0.05 we report the comparison as consistent in direction and not statistically established. Second, the Porto Seguro and UCI credit win rates are driven by the twenty partitions added in the extension (hypervolume 18/20 and 19/20 against 6/10 and 5/10 with one tie in the first ten; on UCI credit the batch difference is itself significant under an identical protocol, Fisher $p = 0.009$): the effect is real in direction, but its magnitude varies enough between draws of ten partitions that ten were not enough to see it. Third, it depends on the cost definition: under the support cost Santander is unchanged (30/30) and Porto Seguro stays positive on IGD^+ (29/30), but BNP Paribas weakens to 19–20/30 and UCI credit *reverses*, NBI-B being worse in 26–27/30 partitions (full support-cost repetition in supplementary Table S1).

How much of this is the relocated geometry? Because a vertex value of β returns its anchor directly, NBI-B’s returned set *contains* the three real single-objective optima, which are also points of the empirical reference. Part of the gain is therefore the injection of those extreme solutions rather than the relocated CHIM. We separate the two by rescoreing NBI-A’s own candidate set augmented with the same three anchors, changing nothing else (`scripts/pco213_anchor_injection_contr`). Injecting the anchors alone closes a median 80% of the A-to-B hypervolume gap on Santander, 90% on BNP Paribas, 49% on Porto Seguro and 1% on UCI credit. The residual, which is the part attributable to the relocated geometry, is +0.038, +0.005, +0.068 and +0.097 hypervolume respectively.

This materially qualifies the headline. On the two datasets with the largest raw effect the mechanism is mostly the returned extreme points, not the CHIM; on UCI credit it is almost entirely the CHIM; Porto Seguro is intermediate. The practical recommendation is unchanged and if anything simpler—compute the real single-objective optima and make sure they reach the returned set, whether by re-anchoring or by adding them—but the causal story must be stated at this resolution rather than as “relocating the CHIM repairs the front”.

What goes wrong when anchors come from the surrogate. NBI-A’s BNP Paribas hypervolume ratio splits into 21 partitions at 0.89–0.93, two intermediate and seven collapsing to 0.04–0.12, and the seven are exactly those in which the parsimony rule selects the *quadratic* ROC-AUC order (7/7 against 0/22 of the linear-selected ones). A quadratic surface moves the surrogate AUC anchor off the gradient-boosting vertex to an interior mixture the empirical reference dominates; in six of the seven that mixture carries k -nearest neighbours at 16–31 ms per 1,000 rows, so anchor cost tracks the collapse (Spearman -0.69), but one partition collapses with a cheap anchor, making cost a marker rather than the mechanism.

The consequence for the gate is uncomfortable. Because the quadratic order is selected precisely when it predicts unseen points better, the collapsed partitions have *higher* external R_{ext}^2 (median 0.46 against 0.08) and pass the gate *more* often (3 of 7 against 2 of 23) than those where NBI-A works: a gate on surface quality does not protect against this failure, while real anchors do, NBI-B reaching 0.935–0.996 in all thirty partitions. It also settles how the gate-conditional analysis must be read — the apparently larger anchor benefit where the gate passes on BNP Paribas (+0.64 against +0.21) is NBI-A’s collapse restated, since three of the five gate-passing partitions are collapse partitions, and is not evidence that real anchors add more when the surface is reliable.

5.4 RQ5: Surrogate NBI against metamodel-free NBI

NBI-C gives the best or tied-best approximation among the surrogate-derived methods on every dataset (Section 5.5 shows an external evolutionary optimizer does better still). NBI-C’s median hypervolume ratio is 0.989, 0.983, 0.982 and 0.976. It exceeds 0.97 in every partition of Santander and BNP Paribas but only in 24/30 and 25/30 partitions of Porto Seguro and UCI credit; uniformly, it exceeds 0.95 in all thirty partitions of three datasets and in 26/30 on Porto Seguro (worst partition 0.877). Against NBI-B it wins on hypervolume in 28, 24, 28 and 30 of 30 partitions, at medians of +0.0084, +0.0106, +0.074 and +0.275, the corrected test reaching significance on Santander ($p = 0.003$) and UCI credit ($p = 0.010$).

Set beside the anchor effect of Section 5.3, these medians do not order the two steps the same way on the four datasets. Removing the metamodel adds far less than moving the anchors on Santander (+0.0084 against +0.196 hypervolume) and on BNP Paribas (+0.0106 against +0.058); on Porto Seguro the two are a near-tie on the median (+0.074 against +0.076) and reverse on the mean (+0.108 against +0.085); and on UCI credit the ordering reverses outright, removal adding three times what the anchors add (+0.275 against +0.092; on the mean, +0.277 against +0.060). Two clean datasets, one tie and one reversal over four datasets is an observation about these problems, not a tested ordering, and it is weakened further by the confound of

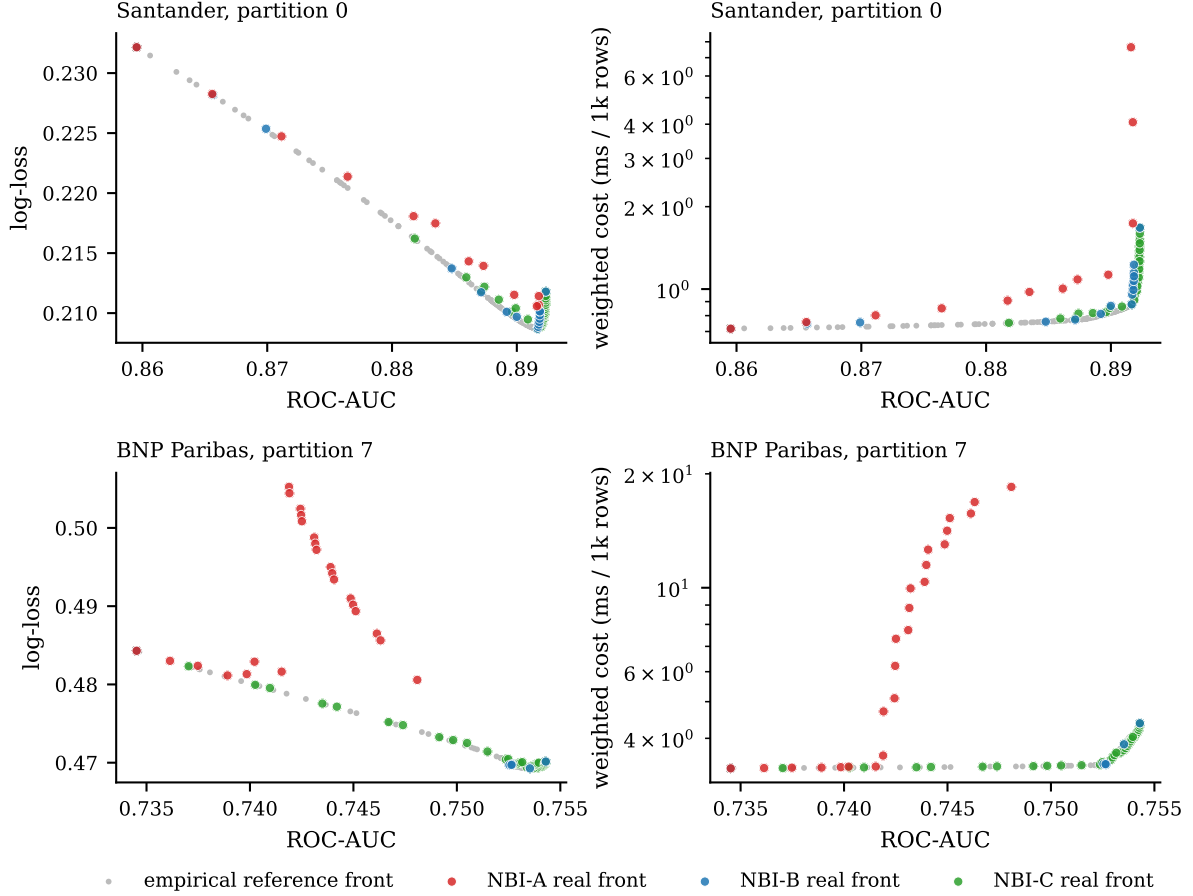


Figure 6: Revalidated fronts of the three NBI variants against the empirical Pareto reference, in the accuracy–calibration and accuracy–cost projections, for one Santander partition (top) and one BNP Paribas collapse partition (bottom). Surrogate anchors displace the NBI-A front away from the reference and up the cost axis.

Section 3.4: NBI-C differs from NBI-B in the solver acceptance rule as well as in the objective source, so the *ratio* of the two effects is weaker evidence than either effect on its own.

Three caveats bound the NBI-C result. On BNP Paribas the two endpoints disagree, but the disagreement is a property of the reference definition rather than of the arms. Against the augmented reference of Table 4, NBI-C covers more volume (24/30, median +0.0106) while NBI-B is closer in IGD^+ (24/30, median −0.0028); against the method-independent sample core the IGD^+ comparison reverses — NBI-C is closer in 20/30, median +0.0025 — while the hypervolume direction is unchanged (24/30, median +0.0077) (`nsga2/nsga2_pareto_quality.csv`). We therefore withdraw the earlier reading of this as a genuine hypervolume/ IGD^+ trade-off between the arms and record it as reference-dependent. The large UCI credit gap is not a surrogate-accuracy effect — those surfaces pass the gate in every partition — but a convergence effect: NBI-B certifies a median of 29 of its 66 subproblems, indicators use converged candidates only, and the gap correlates with that count at Spearman −0.92; retaining NBI-B’s unconverged iterates preserves the direction (30/30) but cuts the mean gap from +0.277 to +0.108, and in the three partitions where NBI-B converged fully the gap is at most 0.01. Neither observation supports a claim that interior surrogate error dominates once anchors are right. Third, the premium is large and uneven (Table 5): NBI-C consumes about 4.2×10^5 real objective evaluations per replication where the surrogate arms consume none. Charged standalone, so that each arm carries the stages it depends on (Table 5), the premium of NBI-C over NBI-B is 6.2, 5.9, 5.9 and $2.3\times$ — smallest on UCI credit, where it buys the largest gain, and 73-fold on Santander for

Table 5: Standalone cost of each NBI variant: mean wall-clock seconds per partition (Apple M4 Max, 8 worker threads) and mean real out-of-fold objective evaluations, charging every arm for everything it needs to run on its own. NBI-A is charged the 66-run design, the 100 gate compositions and the Scheffé fit (“design + fit”), since its anchors are the optima of the fitted surfaces; NBI-B is charged the same *plus* the single-objective reference stage, which is where its real anchors come from and which is dominated by the direct ROC-AUC search over 4×10^4 sampled compositions; NBI-C is charged the reference stage and its own solve, and needs neither the design nor the surfaces. Real anchors are therefore not free: NBI-B costs $1.2\text{--}12.7\times$ the standalone wall clock of NBI-A, and the standalone premium of NBI-C over NBI-B is $6.2, 5.9, 5.9$ and $2.3\times$ (Santander, BNP Paribas, Porto Seguro, UCI credit), not the solver-stage ratio. Charging the whole reference stage to NBI-B and NBI-C is an upper bound, since that stage also computes references the arms do not consume, and the SLSQP log-loss anchor’s evaluations are not separately instrumented. Not charged to any arm: the shared out-of-fold stage (335, 86, 69, 11 s) and the study’s own instrumentation — empirical reference, quality scoring and comparators (270, 154, 302, 74 s). “Success” is the mean fraction of the 66 subproblems that terminated successfully (A/B: SLSQP-certified; C: equality-feasible under the lenient rule of Section 3.4). Source: `tables/stage_times.csv`, `tables/nbi_runs.csv`, `references.json` and `design_points.npz` of the frozen replications.

Dataset	Arm	Stage means (s)			Standalone		Real	
		Design + fit	Single-obj. refs	NBI solve	total (s)	vs. NBI-A	evals	Success
Santander	NBI-A	4.1	—	34.0	38.1	$1.0\times$	166	0.98
	NBI-B	4.1	444.5	34.6	483.1	$12.7\times$	4.1×10^4	0.95
	NBI-C	—	444.5	2528.4	2972.9	$78.0\times$	4.5×10^5	0.83
BNP Paribas	NBI-A	2.2	—	23.1	25.3	$1.0\times$	166	0.95
	NBI-B	2.2	243.2	35.6	281.1	$11.1\times$	4.1×10^4	1.00
	NBI-C	—	243.2	1409.8	1653.0	$65.2\times$	4.6×10^5	0.83
Porto Seguro	NBI-A	4.0	—	84.8	88.8	$1.0\times$	166	0.58
	NBI-B	4.0	441.2	55.8	501.0	$5.6\times$	4.1×10^4	1.00
	NBI-C	—	441.2	2504.5	2945.7	$33.2\times$	4.6×10^5	0.98
UCI credit	NBI-A	0.5	—	124.8	125.3	$1.0\times$	166	0.39
	NBI-B	0.5	54.1	99.7	154.4	$1.2\times$	4.1×10^4	0.48
	NBI-C	—	54.1	306.6	360.7	$2.9\times$	4.7×10^5	0.98

+0.008 hypervolume on a dataset whose log-loss surface is already exact. Metamodel-free NBI is the safe default, justified where the surfaces or the anchors fail, not uniformly preferable.

Cheap comparators. Random weighted scalarization is strong where the surfaces are reliable — it ties NBI-C on UCI credit (differences within ± 0.002 , win fractions 16/30 and 18/30, corrected $p = 1.00$), matches it on Santander hypervolume (0.978 against 0.989) and beats it on Santander IGD⁺ in 20/30 — and collapses where they are not, to a median of one front point and hypervolume ratio 0.474 on BNP Paribas. The comparison is confounded and we say so: NBI-C optimizes exact objectives while scalarization optimizes surrogates, so its BNP Paribas and Porto Seguro advantage cannot be attributed to the NBI construction rather than to the objective source. On identical surfaces NBI-B beats scalarization 30/30 on BNP Paribas and 25/30 on Porto Seguro but loses 29/30 on UCI credit.

The 66-point random Dirichlet sample loses to every method everywhere — its hypervolume ratio is exactly zero in 10/30, 21/30 and 25/30 partitions on Santander, BNP Paribas and Porto Seguro, because uniform compositions cost 8 to 20 times more than the reference front, and the unoptimized 66-run mixture design also beats it 30/30 everywhere — but it is matched to the NBI sets only in the number of candidates returned, so this is a floor check and no efficiency claim is made from it.

One negative result bears on the classical motivation for NBI recalled in Section 3.4: the even spread of subproblems over the CHIM does not survive revalidation on the real objectives, under any spacing definition we computed. NBI-C’s size-matched spacing percentile is 0.99, 0.54,

Table 6: Evaluation-matched NSGA-II against metamodel-free NBI (NBI-C), weighted cost, scored against the common augmented reference. Both optimize the same three real out-of-fold objectives and consume a matched number of real objective evaluations per replication (realized ratio 0.999996 overall, 0.99992–1.00010 per run). Δ is the paired difference signed so that $\Delta > 0$ favours NSGA-II, with its percentile-bootstrap 95% interval, the win/tie/loss count over 30 partitions and the Holm-corrected Nadeau–Bengio p . The last column is the wall-clock ratio, which is *not* matched. The direction is the same under the sample-core reference (supplementary Table S2). Source: [reports/pco213_postwork_benchmark/nsga2/](#).

	Median HV ratio		Δ IGD ⁺				Δ HV ratio				Time
Dataset	NBI-C	NSGA-II	median [95% CI]	W/T/L	p	median [95% CI]	W/T/L	p	ratio		
Santander	0.990	0.998	+0.0060 [+0.0039, +0.0074]	29/0/1	0.026	+0.0083 [+0.0063, +0.0110]	30/0/0	0.026	2.2×		
BNP Paribas	0.981	0.994	+0.0087 [+0.0078, +0.0094]	30/0/0	<0.001	+0.0127 [+0.0113, +0.0139]	30/0/0	<0.001	2.2×		
Porto Seguro	0.981	0.995	+0.0048 [+0.0035, +0.0075]	26/0/4	0.577	+0.0115 [+0.0070, +0.0169]	26/0/4	0.577	2.2×		
UCI credit	0.977	0.988	+0.0032 [+0.0011, +0.0045]	23/2/5	0.131	+0.0114 [+0.0071, +0.0156]	30/0/0	0.023	2.3×		

0.96 and 0.41, against 1.00, 1.00, 1.00 and 0.93 for the random Dirichlet sample; Section 5.5 shows this is a property of the construction, not of the objective geometry.

5.5 RQ5b: An evaluation-matched evolutionary baseline

Everything above compares methods within one methodological family; the external question is whether a canonical evolutionary optimizer, given the same objectives and budget, behaves differently. We ran NSGA-II (Deb et al., 2002) in pymoo (Blank and Deb, 2020) on exactly the three real out-of-fold objectives NBI-C optimizes, with a population of 66 matching the 66 NBI directions, library-default variation operators, a deterministic simplex repair, and a termination budget set *per replication* to the real objective evaluations NBI-C consumed there; the configuration was fixed and committed before any run, so no operator setting was chosen by inspecting performance (supplementary Section S13). Across the 120 replications the realized budget ratio is 0.999996 overall and lies in [0.99992, 1.00010] per run, so the comparison is genuinely evaluation-matched. It is not time-matched, and the wall-clock ratio is reported separately. Because the empirical reference is augmented with candidate sets, adding a method changes it; we therefore score twice, against a *sample-core* reference containing the search output of no optimizer, and against a *common augmented union* that includes NSGA-II and against which every method’s indicators are recomputed.

The answer is unambiguous and we state it plainly: **NSGA-II attains a better approximation than NBI-C on both primary endpoints, on all four datasets, under both reference definitions** (Table 6; the sample-core repetition is supplementary Table S2). All 16 dataset $\times$ endpoint $\times$ reference cells favour NSGA-II, with rank-biserial correlations between +0.87 and +1.00. The corrected test reaches significance on Santander, BNP Paribas and UCI credit hypervolume; on Porto Seguro the paired differences are heavy-tailed (mean +0.021 against median +0.011), so by the rule fixed in advance the median, the win fraction and the effect size are the evidence there, and all three agree.

Four qualifications matter. The margins are small in absolute terms: median hypervolume-ratio gains of +0.006 to +0.013, on a scale where NBI-C already reaches 0.977–0.990. The comparison is evaluation-matched but not cardinality-matched: NSGA-II returns exactly 66 mutually non-dominated points in all 120 runs, against NBI-C revalidated front sizes whose medians are 35, 34, 56 and 65. Hypervolume is monotone under set inclusion and IGD⁺ improves with additional points near the reference, so the larger returned set mechanically helps NSGA-II on both endpoints, and within datasets the per-partition cardinality gap correlates with the hypervolume-ratio gap at Pearson +0.50, −0.37, +0.57 and +0.68. That bounds the margin rather than removing it — the hypervolume gap is +0.0114 on UCI credit, where the two sets differ by a median of one point, and the BNP Paribas correlation is negative — but the reported

Table 7: Holdout ranking agreement (partitions in which the OOF-best knee pick among NBI-A/B/C and scalarization is also holdout-best, weighted cost) and cost-definition sensitivity (partitions in which the hypervolume-best set differs between the weighted and the support cost; median Spearman correlation of the set rankings). Wilson 95% intervals. Sources: `statistics/proportion_intervals.csv`, `statistics/cost_definition_sensitivity_r30.csv`.

Dataset	Holdout ranking agreement	Best set differs (W vs S)	Rank ρ (W vs S)
Santander	30/30 [0.89, 1.00]	10/30 [0.19, 0.51]	0.83
BNP Paribas	15/30 [0.33, 0.67]	24/30 [0.63, 0.90]	0.89
Porto Seguro	23/30 [0.59, 0.88]	8/30 [0.14, 0.44]	0.94
UCI credit	30/30 [0.89, 1.00]	20/30 [0.49, 0.81]	0.66

margins should be read as an upper bound on NSGA-II’s advantage. NSGA-II costs about $2.2\times$ the wall clock of NBI-C at equal evaluation count, because a generation of 66 evaluations amortizes fixed costs less well than NBI’s batched subproblem solves. And NSGA-II contributes 8–29% of the augmented reference front, so like NBI-B and NBI-C it is partly graded against its own output there — which is why the sample-core reference, where it contributes nothing, is reported alongside and gives the same answer.

This also settles what the surrogate-only comparison could not: whether NBI’s lost spacing advantage reflects the objective geometry or the CHIM construction. NSGA-II’s crowding-distance selection produces markedly more evenly spaced revalidated fronts on every dataset — median spacing coefficient of variation 0.74, 0.62, 0.60 and 0.65 against NBI-C’s 2.28, 0.88, 1.56 and 1.92 — so even spacing here is attainable and CHIM-based construction does not deliver it. Scored post hoc under the support cost, which NSGA-II never optimized, the advantage largely transfers (supplementary Table S2), most strongly on BNP Paribas; the one reversal is Santander IGD⁺, where NBI-C is marginally closer in 17 of 30 partitions.

This bounds one claim — NBI-C is best or tied-best *among the surrogate-derived methods*, not among multiobjective optimizers generally — and leaves untouched every conclusion about the surrogate pipeline itself (reliability, anchor misplacement, the gate’s blind spot, the coefficient result, the cost-definition result), because NSGA-II is not part of that pipeline and was never proposed as a component of it. A conventional optimizer on the same objectives behaves differently, modestly and consistently, and a practitioner whose evaluations are cheap enough to run either should prefer it.

5.6 RQ6: The actual multiobjective conflict

In objective space the conflict between discrimination and calibration is almost degenerate: the direct ROC-AUC optimum exceeds the SLSQP log-loss optimum by 0.00068, 0.00080, 0.00006 and 0.00034 ROC-AUC, with bootstrap interval half-widths below 0.00004 — 5 to 100 times less than the gain from ensembling at all. Two qualifications prevent the stronger statement that the problems are the same: the optima differ in weight space (L_1 distances 0.35, 0.38, 0.06 and 0.20, the Porto Seguro pair practically coinciding), and Santander pays 0.0032 of log-loss for its 0.00068 of ROC-AUC, so the practitioner gets a different *model* even where the objective values are nearly indistinguishable. An apparent +254 ms per 1,000 rows deployment-cost gap on Santander did not survive checking and we do not claim it: it is an artifact of the activity threshold in Equation (6), a near-threshold k -nearest-neighbour weight that is AUC-inert, and an AUC-equivalent blend without k -nearest neighbours costs about +4 ms per 1,000 rows (supplementary Section S9).

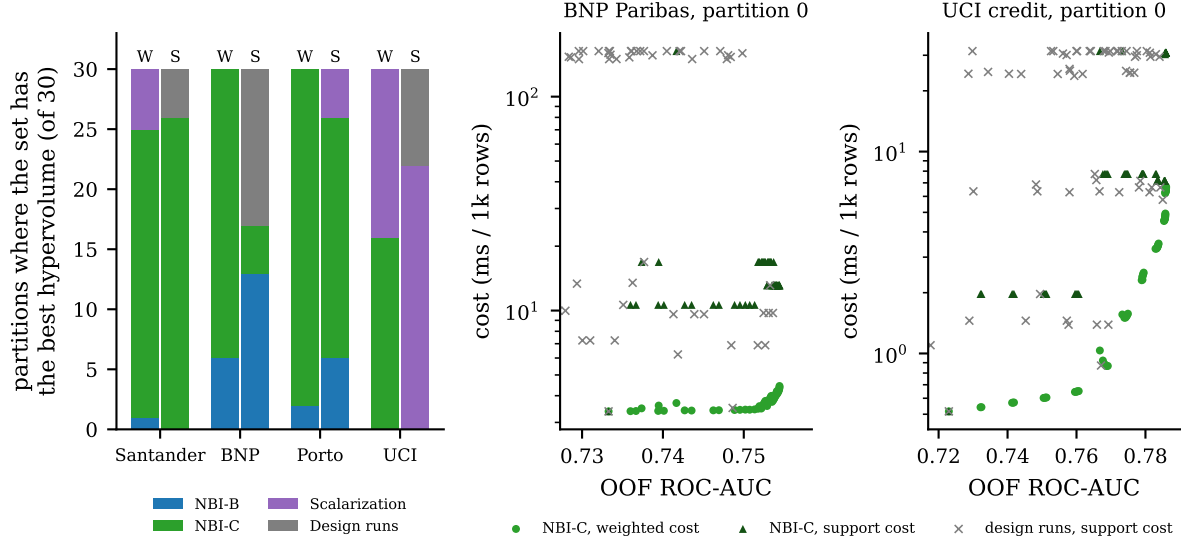


Figure 7: Left: which method attains the best hypervolume, per dataset, under the weighted cost (W) and the support cost (S), over 30 partitions. Right: one NBI-C candidate set scored under both definitions against the raw design runs, for one BNP Paribas and one UCI credit partition. Under the step cost the design runs occupy cheap supports the NBI sets, optimized on the linear relaxation, do not reach.

5.7 RQ7: Effect of the cost definition

The two cost definitions disagree often enough to change conclusions (Figure 7, Table 7): the hypervolume-best method changes between them in 10/30 partitions on Santander, 24/30 on BNP Paribas, 8/30 on Porto Seguro and 20/30 on UCI credit. NBI-C remains best under both on Santander and Porto Seguro (26/30 and 20/30); on UCI credit random scalarization takes the support-cost hypervolume in 22/30, and several paired readings reverse outright (Section 5.3, Section 5.4).

The mechanism, where it can be isolated, is that a method optimizing $\sum_i w_i c_i$ has no incentive to zero out a component. The raw 66-run design occupies exactly the cheap supports the NBI sets miss and can win outright under a step cost — as it does in 13/30 BNP Paribas partitions.

One earlier reading did not survive scrutiny and we correct it here: the apparent bimodality of NBI-C’s BNP Paribas support hypervolume is a normalization artifact, not a regime. The per-partition normalization box has a cost bound near 17 ms per 1,000 rows in the 13 partitions where no k -nearest-neighbour-bearing point survives on the support reference front and near 160–167 in the other 17 — a tenfold rescaling of one axis — while NBI-C’s fronts are the same in both groups, and under a common bound the split disappears. We therefore claim only the qualitative statement: NBI sets optimized on the linear cost omit the cheapest supports the raw design contains, and the penalty’s size depends on the chosen cost range.

5.8 RQ8: Transfer to the untouched holdout

In level, yes. The mean holdout-minus-out-of-fold ROC-AUC of every knee-selected solution lies within ± 0.005 as a point estimate, and the shift equals the weight-averaged shift of the base models on the same partition (correlation 0.97–1.00): a partition and refitting effect shared with the constituent models, not selection optimism.

In ranking, it depends on the dataset. The out-of-fold-best of the four knee picks is also holdout-best in 30/30 partitions on Santander, 15/30 on BNP Paribas, 23/30 on Porto Seguro and 30/30 on UCI credit, and the twenty added partitions reproduce this pattern exactly (20/20,

Table 8: Robustness from $R = 10$ to $R = 30$: mean hypervolume ratio and IGD^+ (weighted cost) of each NBI variant estimated from partitions 0–9 (the frozen $R = 10$ study, tag `pco213-postwork-r10`) and from all 30 partitions, with the $R = 30$ bootstrap interval and whether the $R = 10$ estimate lies inside it. Source: `statistics/r10_vs_r30_stability.csv`.

Dataset	Set	Hypervolume ratio			inside	IGD^+			inside
		$R = 10$	$R = 30$ [95% CI]			$R = 10$	$R = 30$ [95% CI]		
Santander	NBI-A	0.846	0.800 [0.767, 0.833]		no	0.0954	0.1321 [0.1078, 0.1573]		no
	NBI-B	0.982	0.980 [0.977, 0.982]		yes	0.0085	0.0121 [0.0101, 0.0142]		no
	NBI-C	0.991	0.987 [0.985, 0.990]		no	0.0069	0.0105 [0.0084, 0.0127]		no
BNP Paribas	NBI-A	0.717	0.695 [0.560, 0.819]		yes	0.2212	0.2502 [0.1234, 0.3877]		yes
	NBI-B	0.966	0.974 [0.969, 0.978]		no	0.0138	0.0114 [0.0096, 0.0132]		no
	NBI-C	0.982	0.982 [0.981, 0.984]		yes	0.0143	0.0142 [0.0134, 0.0151]		yes
Porto Seguro	NBI-A	0.803	0.781 [0.750, 0.812]		yes	0.0751	0.0911 [0.0765, 0.1055]		no
	NBI-B	0.820	0.866 [0.810, 0.913]		yes	0.0695	0.0531 [0.0351, 0.0746]		yes
	NBI-C	0.973	0.974 [0.964, 0.983]		yes	0.0113	0.0111 [0.0075, 0.0155]		yes
UCI credit	NBI-A	0.661	0.639 [0.589, 0.693]		yes	0.1854	0.2043 [0.1724, 0.2340]		yes
	NBI-B	0.670	0.699 [0.643, 0.751]		yes	0.1809	0.1681 [0.1383, 0.2008]		yes
	NBI-C	0.975	0.976 [0.974, 0.979]		yes	0.0107	0.0099 [0.0089, 0.0110]		yes

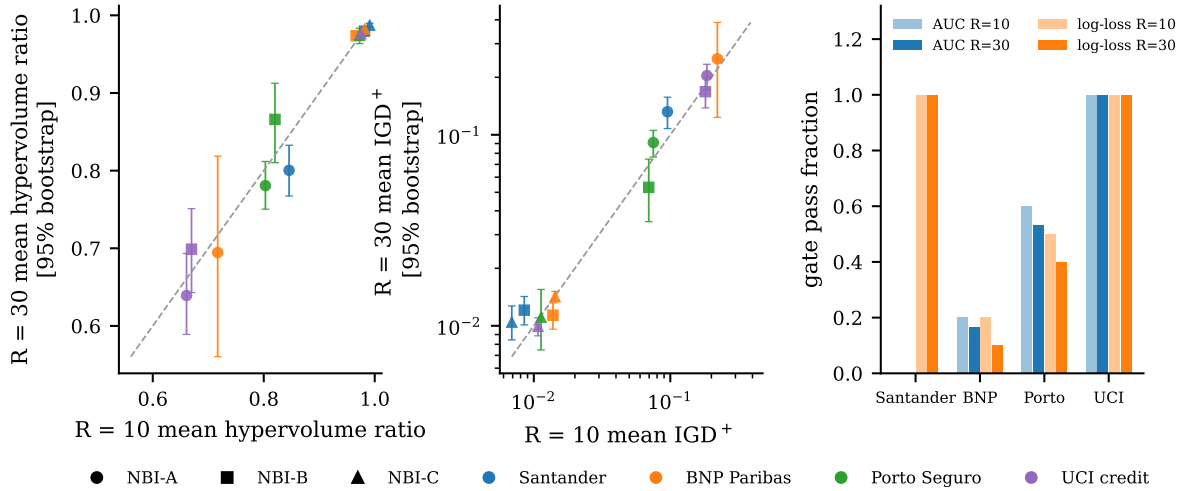


Figure 8: Mean indicator at $R = 10$ against $R = 30$ with bootstrap intervals (left, centre) and gate frequencies at both sample sizes (right). Points on the diagonal are unchanged estimates.

10/20, 15/20, 20/20). The BNP Paribas figure measures how close the four candidates are, not a failure of selection: their median out-of-fold gap is 0.0005 against a holdout noise standard deviation of about 0.0012, and when the two disagree the holdout regret is at most 0.0022 ROC-AUC. These rates are specific to the selection rule and cost definition — under the support cost they become 28, 28, 17 and 16 of 30 — and are not dataset properties.

5.9 RQ9: Did ten replications suffice?

Tripling the number of replications changed no directional conclusion (Table 8, Figure 8): the $R = 10$ estimate lies inside the $R = 30$ bootstrap interval in 9 of 12 hypervolume cells, 7 of 12 IGD^+ cells and all 8 gate frequencies, and the sign of the median paired effect is identical in all 32 primary cells. What the extension delivered is a change of status for three findings — the Porto Seguro and UCI credit anchor effect went from an undecidable 6/10 and 5/10 to 24/30 with intervals excluding zero; the BNP Paribas anchor collapse became a quantified rate (7/30) with an identified trigger (quadratic order selection); and the BNP Paribas support-cost bimodality

and the Santander deployment-cost gap were exposed as artifacts. A fourth item on that list has since been withdrawn: the NBI-C versus NBI-B hypervolume/IGD⁺ split on BNP Paribas is a property of the reference definition and reverses under the sample core (Section 5.4), so the extension sharpened that estimate without settling a finding. The criterion is itself lenient, because partitions 0–9 are one third of the $R = 30$ sample: against the twenty new partitions alone only 6 of 12 hypervolume and 5 of 12 IGD⁺ cells agree, and the Porto Seguro and UCI credit anchor win rates differ between batches under an identical protocol. That disagreement is the magnitude of partition sensitivity the replication was designed to measure (supplementary Section S11).

6 Discussion

One claim organizes the results: on the classifier-weight simplex the mixture-design response surface is a good *description* of ensemble performance and an unreliable *optimization substrate*, and the dominant error is not inside the surface but in the points that build the geometry of the search.

6.1 Why anchors dominate

NBI derives its whole search geometry from the payoff matrix, so every subproblem in Equation (12) is posed relative to the individual optima. An anchor error is therefore global rather than local: it relocates the CHIM, redirects every quasi-normal and rescales the objective space of all 66 subproblems, while an interior error of the same size perturbs single solutions and leaves the frame intact.

The asymmetry predicts part of what we observe, but not all of the measured gap is geometry. Replacing surrogate anchors with real ones, surfaces held fixed, improves both primary endpoints in 30/30 partitions on the two datasets whose ROC-AUC surfaces almost never pass the gate, and in 23–24/30 on Porto Seguro and 24/30 on UCI credit, with hypervolume gains up to +0.18. That substitution changes four things at once (Section 3.4), and one of them is not geometric: because the three vertex values of β return their anchor directly, NBI-B’s candidate set *contains* the real single-objective optima, which are themselves points of the empirical reference. The injection control of Section 5.3 separates that channel from the rest. Adding those same three points to NBI-A’s own candidate set, changing nothing else, already closes a median 80% of the A-to-B hypervolume gap on Santander and 90% on BNP Paribas, against 49% on Porto Seguro and 1% on UCI credit. On Santander and BNP Paribas — the two datasets with the largest raw effect, and the first of them the only Holm-significant one — most of what the headline numbers measure is therefore the returned extreme solutions rather than the relocated CHIM. The relocated-CHIM effect is cleanest on UCI credit, where it accounts for essentially the whole +0.097 hypervolume residual; it is present but small elsewhere, positive in 30/30 Santander and BNP Paribas partitions at a median +0.038 and +0.005. The mechanism therefore survives at a smaller size than the raw contrast suggests, and the practical recommendation is if anything simpler: compute the real single-objective optima and make sure they reach the returned set, whether by re-anchoring or by adding them.

How much the surrogate still costs once the anchors are right is dataset-dependent, and no single ordering covers the four. Removing the metamodel entirely adds far less than real anchors did on Santander (median +0.008 hypervolume against +0.196) and on BNP Paribas (+0.011 against +0.059); on Porto Seguro the two effects are the same size on the median (+0.074 against +0.076) and reverse on the mean (+0.108 against +0.085); and on UCI credit the ordering reverses outright (+0.275 against +0.092). Two clean cases, one tie and one reversal over four datasets make that an observation rather than a tested claim, and because B versus C also changes the solver acceptance rule (Section 3.4), the *ratio* of the two effects is weaker

evidence than either effect on its own.

BNP Paribas shows how the anchors go wrong in the first place: where the parsimony rule selects a quadratic ROC-AUC surface, the surrogate maximum sits at an interior mixture instead of the gradient-boosting vertex, so the CHIM is built from a false utopia point and the front collapses (Section 5.3) — a geometric failure, not a numerical one, since the solver certifies these subproblems, and invisible to any diagnostic computed on the surface alone.

6.2 Why the reliability gate does not catch it

The gate asks whether a surface predicts unseen compositions well on average; anchor placement depends on where the surface puts its *argmax*, a far more local property. A surface can be globally accurate and still misplace an extremum, or globally poor with its *argmax* on the right vertex.

On BNP Paribas they diverge actively: the parsimony rule selects the quadratic order precisely when it predicts held-out points better, so the collapse partitions have higher external R^2 and pass the gate more often than the partitions where the surrogate anchor is right. A linear Scheffé surface has its maximum at a vertex by construction, so the *less* flexible model is accidentally protected while the more flexible one may place an interior maximum the data do not support.

Hence a division of labour: the gate is worth its 100 extra evaluations for separating unusable surfaces from excellent ones, but only the cheaper anchor check that distinguishes NBI-B from NBI-A removes the failure instead of diagnosing it.

6.3 Smooth surrogates against a rank statistic

Two of the three objectives behave as the framework’s engineering origins assume: log-loss is convex and smooth in $\mathbf{w}$, and the Brier score is exactly quadratic in $\mathbf{w}$, so its surface reproduces it at $R_{\text{ext}}^2 = 1.000$ in all 120 replications — an algebraic identity that checks the fitting code, not evidence about adequacy. The ROC-AUC is not: a rank statistic, constant wherever the induced ordering does not change and kinked between orderings, which a low-order polynomial smooths across instead of representing. Hence the Santander ROC-AUC surface reaches a Spearman correlation of 0.830 with a negative external R^2 — ordering broadly right, values wrong.

The resulting bias is one-directional (Section 5.2): a quadratic fitted over an edge with one weak endpoint absorbs that curvature into β_{ij} and overshoots its maximum — the same over-prediction of interior optima that displaces surrogate anchors off vertices. Where the objective *is* smooth in the weights the machinery can work as advertised: Santander’s log-loss surface passes the gate in every partition.

Non-smoothness is a contributing factor, not the explanation. Were it the explanation, the smooth convex objective would be the reliable one everywhere, and it is not: log-loss surfaces pass the gate in 30/30 partitions on Santander and UCI credit but in only 3/30 on BNP Paribas and 12/30 on Porto Seguro. On BNP Paribas the smooth objective does *worse* than the rank statistic (3/30 against 5/30; median external R_{ext}^2 of -0.180 against $+0.118$), and its external RMSE of 0.121 nats is 17.7% of the response range, the worst relative external fit anywhere in this study; on Porto Seguro the two are effectively tied (12 against 16). Only Santander shows the sharp separation the smoothness account predicts (R_{ext}^2 0.971 for log-loss against -0.333 for ROC-AUC), and one dataset is an illustration rather than a mechanism. What these data establish is that adequacy varies by dataset *and* by metric and has to be measured on unseen compositions; they do not identify what, besides the shape of the response in $\mathbf{w}$, makes a given dataset–metric pair fittable by a low-order polynomial.

6.4 Non-smoothness also constrains the optimizer

The same property constrains the metamodel-free arm: SLSQP cannot certify optimality on a piecewise-constant objective, whose gradient is zero almost everywhere, so NBI-C accepts a

feasible iterate under the weaker rule of Section 3.4, not the certification NBI-A and NBI-B obtain on smooth polynomials. Success rates are therefore not on a common scale across arms, and we do not read the difference as evidence about surrogate quality.

UCI credit shows why that care is needed: its +0.277 hypervolume gain for NBI-C over NBI-B, at Holm-corrected $p = 0.010$, looks like the strongest evidence here for the metamodel-free arm, yet UCI credit’s surfaces pass the gate in every partition, so surrogate error cannot explain it: NBI-B certifies a median of 29 of its 66 subproblems, the gap correlates with the converged count at Spearman -0.92 , and in the three partitions where NBI-B converges fully it disappears.

6.5 Compute against fidelity

Metamodel-free NBI is best or tied-best on the primary endpoints, so it is a defensible default, but it spends about 4.2×10^5 real evaluations per replication against none for the surrogate arms, at 2.3 to 6.2 times the wall-clock cost. That spend is strongly conditional: on UCI credit the threefold premium buys the largest gain, while on Santander a seventy-three-fold premium buys +0.008 hypervolume on a dataset whose log-loss surface is already exact.

Its affordability is specific to this problem: one evaluation is a matrix–vector product plus a sort, 0.37 to 2.13 ms here. Where each evaluation needs a model refit the ordering reverses and the surrogate is the only viable route — the regime the framework was designed for. The recommendation is therefore conditional on evaluation cost: cheap evaluations should be used directly, at least for the anchors; expensive ones argue for an externally validated surrogate, anchored on whatever real evaluations are affordable.

6.6 Response surfaces as descriptions rather than optimizers

The surfaces themselves are reproducible across resampled partitions (Section 5.2), and as a description of which vertices are weak and which regions flat the model is interpretable in a way a black-box optimizer is not. What does not survive is the step from that description to inference about blends: the classical synergism criterion $\beta_{ij} > |\beta_i - \beta_j|$ is correct as stated and guards against reading the sign of β_{ij} alone, but it is computed *from the surface*, so on these ensembles it inherits the surface’s systematic over-prediction of interior optima, and a candidate pair must be confirmed on the real objective — one matrix product.

6.7 Deployment cost is not the cost that was optimized

Every method here minimized the continuous relaxation $\sum_i w_i c_i$, since a step cost is not differentiable and cannot enter a gradient-based formulation; deployment pays the step cost. The two are not monotonically related inside the simplex, and the consequence is not cosmetic: which of the six candidate sets attains the best hypervolume changes between the definitions in 10, 23–24, 8 and 20 of 30 partitions (Section 5.7), and several paired conclusions reverse, including the anchor effect on UCI credit.

The mechanism is that a linear cost gives no incentive to zero a component: a weight of 0.01 on a model costing 200 ms adds 2 ms under the relaxation and 200 ms in deployment. Methods optimizing the relaxation drift into the interior and miss the cheap low-support solutions — pure vertices and two-model faces — that the unoptimized 66-run design contains by construction.

Whether a relaxation and the support it induces agree has been examined in one related setting, and the answer went the other way: for budgeted learning with feature re-use, a support-cost programme is solved exactly by its linear relaxation (Nan et al., 2015) — a guarantee our setting does not inherit. Report which cost model is used; zero weights below the activity threshold and re-evaluate, since an inert weight just above ε produced a spurious 250 ms difference in our own analysis; and optimize over supports explicitly where deployment cost is the real objective.

6.8 What the framework’s transfer teaches

The pipeline transfers where the response is smooth in the mixture components, a condition identifiable in advance. Where the ROC-AUC surface is adequate too — UCI credit, which passes in 30/30 — the cheap route is competitive: random scalarization on those externally validated surfaces is statistically indistinguishable from the metamodel-free arm (median hypervolume ratio 0.978 against 0.976; mean $\Delta\text{HV} = +0.0015$ with bootstrap interval $[-0.003, 0.007]$, better in 16/30). Surrogate-assisted NBI with real anchors is not — its median ratio there is 0.706 — but that residual gap is a subproblem-convergence effect of NBI-B rather than surrogate error (Section 6.4). The stronger reason to stop early on UCI credit is that there is very little to win: the out-of-fold gain of the log-loss optimum over the best single model is $+0.0007$ ROC-AUC and only $+0.0005$ on the untouched holdout, better in 23 of 30 partitions at an overlap-corrected $p = 0.24$, against holdout gains of $+0.0076$, $+0.0038$ and $+0.0053$ (in 30/30, 30/30 and 29/30) on the other three datasets. What UCI credit therefore shows is not that the inherited machinery is efficient but that on that dataset *no* weighting method separates from the best single model on held-out data, so the cheapest adequate one suffices. The pipeline does not transfer unexamined to a rank-statistic response, to anchors that are vertices of the feasible region rather than interior points, or to an objective that is a step function of the support.

One connection this study does not establish is between front quality and the decision taken from the front. Section 5.8 shows that out-of-fold selection *within* a candidate set transfers to the holdout in level, and that the out-of-fold-best of the four knee picks is also holdout-best in 30/30, 15/30, 23/30 and 30/30 partitions. The cross-method comparison is much weaker. Comparing the knee-selected solutions of two arms directly, paired by partition under the weighted cost, the two contrasts that carry our only Holm-significant primary results have *worse* single-objective holdout accuracy at the selected point: Santander NBI-B minus NBI-A is -0.0047 holdout ROC-AUC, better in 15 of 30 partitions, with holdout log-loss worse in 16 of 30; UCI credit NBI-C minus NBI-B is -0.0119 , better in 6 of 30, with log-loss worse in 24 of 30. Both selected points are also materially cheaper (0.97 and 2.84 ms per 1,000 rows), so these are different points of the same trade-off rather than a contradiction of indicator results computed over whole fronts. They do not support the converse either: on these four datasets a better approximation of the front does not reliably buy a better selected solution, and the contrasts where the pick does improve (BNP Paribas NBI-B minus NBI-A, $+0.0029$ in 28/30; Santander NBI-C minus NBI-B, $+0.0016$ in 26/30) are the ones with the smaller indicator effects. Any statement that out-of-fold selection transfers must therefore be read as within-method and in level, not as evidence that better fronts give better picks.

What we would carry forward is not a method but a checklist: external validation on unseen compositions, an argmax check against the real optimum, revalidation on the true objective before scoring, a reference whose sampled core is constructed independently of the surrogate and of every candidate method, and an explicit statement of which cost is paid. Each of them changed a conclusion that would otherwise have been reported.

7 Limitations

Four datasets is the binding constraint. The statistical unit is one outer partition, but the unit of *generalization* is the dataset, and there are four. Every replication-level interval here describes partition sensitivity within a dataset, not uncertainty about a population of datasets, so every cross-dataset mechanism statement is an association over four points and not a tested hypothesis. The ordering “real anchors matter most where the surrogate is unreliable” is not one we can claim at all: the median anchor effect is largest on Santander, whose ROC-AUC surfaces never pass the gate, and second largest on UCI credit, whose surfaces always pass (Section 5.3). All four datasets are binary tabular problems of 3×10^4 to 2×10^5 rows, and all four are financial-risk problems — retail banking, insurance claims, motor insurance and consumer credit — three

of them from Kaggle competitions, so the four points are not four independent domains. Nothing here establishes behaviour on multiclass problems, text or image data, on very small samples, or on ensembles of more than five members, where the simplex and design grow quickly.

The A-to-B contrast changes four things at once. Replacing the surrogate anchors with the real single-objective optima simultaneously moves the utopia point, rescales both normalization axes through the payoff matrix, reorients the CHIM and all 66 quasi-normal directions, and — because the three vertex values of β return their anchor directly — substitutes three returned candidates that are themselves points of the empirical reference (Section 3.4). It is a contrast between two *specifications* of the anchor source, not a single-factor experiment. The injection control of Section 5.3 separates the fourth of these from the other three by rescaling NBI-A’s own candidate set augmented with the same three anchors, and shows that the split is dataset-dependent: the injected points alone account for a median 80% of the hypervolume gap on Santander and 90% on BNP Paribas, but 49% on Porto Seguro and 1% on UCI credit. It does not separate the first three from one another, and no control in this study does; doing so would require an NBI implementation in which the utopia point, the normalization and the direction set can be varied independently.

The model zoo is fixed and shapes the coefficient findings. Base learners and hyperparameters were fixed in advance with no search: the study measures the weighting layer with the model layer held constant. The zoo deliberately contains a poorly calibrated member and an expensive one, which makes the cost objective non-trivial and also drives the β_{ij} result of Section 5.2: the largest interactions attach to the weakest vertex. A more homogeneous zoo would produce smaller interaction coefficients and a smaller vertex gap to explain, so that finding is conditional on component heterogeneity; these data cannot say how it behaves for a zoo of near-equal members.

The Pareto reference is empirical. With no analytical front, all convergence and coverage indicators are computed against a sample-based reference of about 10^5 points per replication. Residual displacement is small but non-zero: 116 of 120 replications met the 5% tolerance and four stopped at the three-round cap with at most 6%, bounding the objective-space effect at 0.2% of hypervolume. More importantly, the final reference is the non-dominated union of the sampled core with all candidate sets, so methods contributing many points — NBI-B and NBI-C contribute 30–40% — are partly graded against their own output. This inflates their absolute indicator values without affecting the paired contrasts on which every primary claim rests; absolute hypervolume ratios should be read with that in mind.

The three NBI arms are not compute-matched, and NBI-C is not matched to anything. Neither NBI-A nor NBI-B spends real objective evaluations inside the optimizer, but NBI-B is not free: its anchors are produced by the single-objective reference stage (Section 3.5), whose derivative-free ROC-AUC search alone evaluates 4×10^4 compositions per replication. NBI-C consumes about 4.2×10^5 per replication and 2.3 to 6.2 times the wall-clock time of NBI-B; the two cheap comparators consume about 66 evaluations and a few seconds. The comparison is therefore between *methods as specified*, not at equal budget, and supports no efficiency claim. The random Dirichlet comparator is matched only in candidate count and is degenerate under the weighted cost; it is a floor check on the reference geometry. The budget-matched question is answered only at the high end, by the NSGA-II baseline of Section 5.5; the low-budget question — what an evolutionary method or a random-search ladder achieves on the 66 design points and 100 gate points the surrogate arms use — is the one a surrogate paper most needs and the one we do not answer.

Solver semantics differ between the arms. NBI-A and NBI-B optimize smooth polynomials, and their subproblems count as successful only when SLSQP certifies optimality. NBI-C optimizes a piecewise-constant ROC-AUC, where such certification is generically unattainable, so a subproblem is accepted when its equality residual falls below 10^{-3} under a reduced iteration budget. The reported “success rates” are therefore not on a common scale across arms, and the UCI credit gap between NBI-C and NBI-B is substantially a convergence artifact of NBI-B rather than evidence about surrogate quality (Section 5.4); we therefore report both the filtered and unfiltered versions of that comparison.

The support cost is a threshold approximation. Deployment cost is modelled by Equation (6) with $\varepsilon = 10^{-3}$, fixed before the study and never changed. It is a proxy that ignores memory, batching, model loading and shared preprocessing, and treats a weight of 1.1×10^{-3} as a full model and one of 0.9×10^{-3} as free. Section 5.6 shows this is not hypothetical: an AUC-inert k -nearest-neighbour weight just above the threshold produced a spurious 250 ms difference. Any support cost therefore needs a weight-cleaning step, and conclusions that depend on it are sensitive to ε and to the cost range that sets the normalization box.

Cost measurements are single-machine and single-implementation. All c_i are wall-clock medians on one workstation with a fixed library stack and batch size; they would change with hardware, batching, a compiled inference runtime, or an approximate-nearest-neighbour index replacing brute-force search, which would remove the study’s largest cost asymmetry. The cost objective is realistic for the setting measured, not universal.

No claim of universal dominance, and confounded comparators. NBI-C is best or tied-best on the primary endpoints under the weighted cost on these four datasets; it ties with random scalarization on UCI credit, loses to it on Santander IGD⁺, and loses its support-cost advantage on BNP Paribas and UCI credit. Its advantage over scalarization on BNP Paribas and Porto Seguro is confounded with the objective source: NBI-C optimizes exact objectives, scalarization optimizes surrogates. Separating the NBI construction from the objective source would require a scalarization arm on the cached out-of-fold objectives, which we did not run.

Selection rules and the holdout analysis. Holdout transfer is reported for a knee rule under the weighted cost, with TOPSIS and the support cost as sensitivity analyses, and agreement rates differ noticeably between these configurations. They characterize how close the candidates are relative to holdout noise, not the general reliability of out-of-fold selection. What we verify is transfer *within* a candidate set and in level. The cross-method question — whether the arm with the better front also yields the better selected solution — is answered only descriptively, and on the two contrasts that carry our Holm-significant primary results it is answered in the negative at the selected point (Section 6.8); no statement here should be read as “better fronts give better picks”.

Only one external optimizer was evaluated. We added a single canonical evolutionary baseline, NSGA-II at a matched real-objective evaluation budget, and it outperformed the metamodel-free arm on every dataset. That is one algorithm at one configuration on one problem class: the study is not a competition among multiobjective optimizers, and nothing here ranks NSGA-II against MOEA/D, NSGA-III or a surrogate-assisted evolutionary method. Nor is the comparison time-matched — at equal evaluation count NSGA-II costs about $2.2\times$ the wall clock in our implementation, and a differently batched one would shift that ratio.

8 Conclusion

We transferred an established DoE–RSM–NBI framework from engineering and forecasting to classifier ensemble weighting, and asked not whether it produces a Pareto front but when that front can be believed. Answering that required three arms defined by two methodological factors — the source of the objectives and the source of the anchors — revalidation on the true out-of-fold objectives, an empirical reference whose sampled core is constructed independently of the surrogate and of every candidate method, and 4 datasets $\times$ 30 partitions.

The framework is neither uniformly safe nor uniformly unsafe: its behaviour is governed by three properties of the problem, not by anything about the optimizer.

Model class. Surrogate adequacy is a property of the dataset *and* the metric and cannot be assumed from in-sample fit: ROC-AUC surfaces pass an external reliability gate in 0, 5, 16 and 30 of 30 partitions, log-loss surfaces in 30, 3, 12 and 30. Smoothness in the weights is one contributing factor and not the whole explanation. It accounts for Santander, where the rank statistic never passes the gate and has negative external R^2 despite a rank correlation of 0.83 while the convex log-loss passes in every partition; it does not account for BNP Paribas, where the smooth objective does *worse* than the rank statistic (3 of 30 against 5 of 30, median external R^2 of -0.180 against $+0.118$) with the worst relative external fit in the study, or for Porto Seguro, where the two are effectively tied. These data establish that adequacy varies and must be measured on unseen compositions, not what makes a dataset–metric pair fittable.

Geometry. Among failures attributable to the surrogate pipeline, anchor misplacement had the largest observed effect. Replacing surrogate anchors with real single-objective optima improved both primary endpoints in all 30 partitions of Santander and BNP Paribas, whose ROC-AUC surfaces pass the gate in 0 and 5 of 30, in 23 to 24 of 30 on Porto Seguro, which passes in 16 of 30, and in 24 of 30 on UCI credit, which passes in all 30. That substitution changes four things at once, and a control that injects only the three real optima into the surrogate-anchored arm shows how the effect divides: the injected points alone close a median 80% of the hypervolume gap on Santander and 90% on BNP Paribas, but 49% on Porto Seguro and 1% on UCI credit. On the two datasets with the largest raw effect most of what is measured is therefore the returned extreme solutions. The relocated construction — NBI takes its utopia point, normalization and search directions from the payoff matrix, so an anchor error moves the whole frame while an interior error perturbs single subproblems — is the cleanest reading on UCI credit and is positive but small elsewhere. Removing the metamodel altogether, once the anchors are right, adds far less than the anchors did on Santander and BNP Paribas, about the same on Porto Seguro, and three times as much on UCI credit, where a subproblem-convergence effect of the surrogate-anchored arm rather than surrogate error dominates: solver geometry can dominate where the surrogate is excellent. With four datasets, one tie and one reversal, that ordering is an observation and not a tested claim. The first-order fix is in any case the cheapest available: make sure the real single-objective optima a practitioner already computes reach the returned set.

Diagnosis. An external gate on surface quality is worth its 100 extra evaluations and separates usable from unusable surfaces, but it does not predict anchor failure, and on one dataset the two properties diverge: the partitions where surrogate-anchored NBI collapses are exactly those where the parsimony rule selects a surface that fits held-out compositions better and passes the gate more often. Surface accuracy and argmax accuracy are different properties, and only the second governs NBI; the gate should therefore be paired with an anchor check — simply the difference between our two surrogate arms.

Two further results are practical. The classical synergism criterion $\beta_{ij} > |\beta_i - \beta_j|$ is satisfied by the fitted surfaces for the leading classifier pair in every partition of all four datasets, yet the real 50/50 blend beats its better member on only one of them, and across all pairs the surface over-predicts blending gains twenty times and under-predicts none: on these ensembles a large interaction coefficient is a statement about the fit, not evidence of complementarity, so

a candidate pair must be confirmed on the real objective. Separately, the continuous cost a differentiable method can optimize and the step cost deployment pays disagree about which of six candidate sets attains the best hypervolume in 10, 23 to 24, 8 and 20 of 30 partitions across the four datasets (the single-objective references are excluded from that pool), and several conclusions reverse between them. The size of the penalty depends on the cost range that sets the normalization box, so what we claim is the qualitative statement: sets optimized on the linear relaxation omit the cheapest supports the unoptimized design already contains. Any cost claim must name which cost it means, and weight cleaning is necessary — an inert weight just above the activity threshold produced a spurious 250 ms difference in our own analysis before we caught it.

Tripling the replications from 10 to 30 changed no directional conclusion: it converted three undecided findings into quantified ones, exposed two apparent phenomena as artifacts and supplied intervals. We report this because it is evidence about how much replication such a study needs: ten partitions appear to identify the conclusions, while thirty are needed to defend their magnitudes.

One external check bounds the framework’s standing: NSGA-II, given the same three objectives and the same number of real objective evaluations per replication, produced a better approximation than the metamodel-free arm on all four datasets and both endpoints, by small but consistent margins, and markedly more evenly spaced revalidated fronts. Where evaluations are cheap enough to run either, that is the better choice, and the distributed representation NBI is classically credited with is attainable here by a different construction. It does not disturb the diagnosis above — NSGA-II is not part of the pipeline being diagnosed — but it settles what that pipeline is worth against the conventional alternative.

What we carry forward is not a method but a set of controls: validate the surface on compositions it has not seen; compare each surrogate argmax against the real optimum and prefer the real one; revalidate every candidate on the true objective before scoring; build a reference whose sampled core is constructed independently of the surrogate and of every candidate method; and state which cost is being paid. Each is cheap relative to the pipeline it protects, and each changed a conclusion that would otherwise have been reported. The framework remains a good *description* of the ensemble-weight simplex: its coefficients are stable to a few percent across thirty resamplings, and they say something interpretable about which components are weak. Whether it is also a good *optimizer* there depends on conditions that can now be stated in advance and checked cheaply.

Reproducibility statement

Code and frozen states. All code, manifests, aggregated tables, statistics and figures are in the repository `caioribeiro99/doe_nbi_hpo_project`. Two annotated tags freeze the study: `pco213postworkr10` marks the ten-replication analysis and `pco213postworkr30` (commit `b3ed050`) the thirty-replication analysis reported here; every number in this paper is computed from artifacts at that commit, and the comparison of the two tags is Section 5.9. The manuscript, figures and tables are on the branch `paper/surrogatenbiensembleweighting`, cut from that tag.

Pipeline and resumption. The benchmark is driven by a single checkpointed runner (`scripts/pco213_run_postwork_benchmark.py`) with ten stages per replication (out-of-fold estimation, mixture design, Scheffé fitting, single-objective references, empirical Pareto reference, NBI-A, NBI-B, NBI-C, comparators and quality scoring). State is recorded per dataset $\times$ replication $\times$ stage, so an interrupted run resumes without recomputation and never silently recomputes a completed stage. A dry-run mode lists pending work; it was used to verify, before the extension was launched, that replications 0–9 were complete and that exactly 80 replications (800 stages) were pending. Aggregation and statistics are separate scripts (`pco213_postwork_`

`benchmark_report.py` and `..._stats.py`), so the analysis can be rerun without re-executing the experiment. The library has 68 unit tests, including reference implementations checked against scikit-learn, exact hypervolume on known cases, the 66-run design, NBI anchor and warm-start behaviour, and the resumption planner.

Determinism. Every stochastic component is seeded as $20260904 + r$ for replication r : the outer stratified partition, the inner fold assignment, the model random states, the Dirichlet validation compositions (offset +1000 and +2000), the direct AUC search (+3000, +4000), the empirical reference (+5000 onwards) and the comparators (+12000). The seed policy, the full seed map, the configuration history and the resolved library versions are stored in the master manifest.

Environment. Python 3.11.15 with NumPy 2.4.6, pandas 3.0.5, SciPy 1.17.1, scikit-learn 1.9.0, statsmodels 0.15.0 and XGBoost 3.2.0, on macOS (Apple M4 Max, arm64), with eight worker threads. Execution consumed 81.1 hours of stage time (27.5 for the first ten replications, 53.6 for the extension) and produced 3,600 model fits, 19,920 design evaluations, 23,760 NBI subproblems, 5.02×10^7 real objective evaluations, 1.55×10^7 reference points and 4.8×10^6 direct-AUC-search evaluations. Two stages were retried, both in the first ten replications; none in the extension.

Data availability and redistribution limits. The raw datasets are not redistributed. The UCI default-of-credit-card-clients data are downloadable from the UCI Machine Learning Repository; the three Kaggle datasets are subject to competition rules and must be obtained from Kaggle by the user. For each, the repository records the exact source, the acquisition command, the expected row, feature and positive-class-prevalence values and the SHA-256 checksum of the file actually used, so an independent copy can be verified byte-for-byte before use. Where a competition endpoint was unavailable, the file came from a public mirror and was validated against those invariants; the manifests record the source used for each dataset. Competition test labels are never used. Porto Seguro was subsampled to 200,000 rows by a stratified draw with the recorded seed. The per-replication raw artifacts (out-of-fold probability matrices, candidate sets, reference samples) total about 829 MB and are deliberately not version-controlled; they are fully reconstructible from the seeds and the runner.

What a replication requires. Reproducing the aggregate results from the published tables and statistics requires only the repository; from raw data it requires the four datasets, the pinned environment and approximately 81 hours of single-workstation compute, or proportionally less for a subset of datasets or replications, since the runner is resumable and parameterized by dataset and replication count.

Acknowledgements

The computational study was executed on a single workstation; no external compute grant was used.

References

Hussein A. Abbass. Pareto neuro-evolution: Constructing ensemble of neural networks using multi-objective optimization. In *The 2003 Congress on Evolutionary Computation, 2003. CEC '03*, volume 3, pages 2074–2080, 2003. doi: 10.1109/CEC.2003.1299928. Crossref deposits no issued date; OpenAlex records publication year 2004.

- Milad Abolghasemian, Armin Ghane Kanafi, and Maryam Daneshmand-Mehr. Simulation-based multiobjective optimization of open-pit mine haulage system: A modified-NBI method and meta modeling approach. *Complexity*, 2022(1):3540736, 2022. doi: 10.1155/2022/3540736.
- Livio Agnew Bacci, Luiz Gustavo Mello, Taynara Incerti, Anderson Paulo de Paiva, and Pedro Paulo Balestrassi. Optimization of combined time series methods to forecast the demand for coffee in Brazil: A new approach using Normal Boundary Intersection coupled with mixture designs of experiments and rotated factor scores. *International Journal of Production Economics*, 212:186–211, 2019. doi: 10.1016/j.ijpe.2019.03.001.
- Sanghamitra Bandyopadhyay, Sriparna Saha, Ujjwal Maulik, and Kalyanmoy Deb. A simulated annealing-based multiobjective optimization algorithm: AMOSA. *IEEE Transactions on Evolutionary Computation*, 12(3):269–283, 2008. doi: 10.1109/TEVC.2007.900837.
- Gabriela Belinato, Fabrício Alves de Almeida, Anderson Paulo de Paiva, José Henrique de Freitas Gomes, Pedro Paulo Balestrassi, and Pedro Alexandre Rodrigues Carvalho Rosa. A multivariate normal boundary intersection PCA-based approach to reduce dimensionality in optimization problems for LBM process. *Engineering with Computers*, 35(4):1533–1544, 2019. doi: 10.1007/s00366-018-0678-3.
- Julian Blank and Kalyanmoy Deb. Pymoo: Multi-objective optimization in Python. *IEEE Access*, 8:89497–89509, 2020. doi: 10.1109/ACCESS.2020.2990567.
- Oliver Borchert, David Salinas, Valentin Flunkert, Tim Januschowski, and Stephan Günnemann. Multi-objective model selection for time series forecasting. arXiv preprint arXiv:2202.08485, 2022. arXiv:2202.08485.
- Leo Breiman. Stacked regressions. *Machine Learning*, 24(1):49–64, 1996. doi: 10.1007/BF00117832.
- T. G. Brito, A. P. Paiva, J. R. Ferreira, J. H. F. Gomes, and P. P. Balestrassi. A normal boundary intersection approach to multiresponse robust optimization of the surface roughness in end milling process with combined arrays. *Precision Engineering*, 38(3):628–638, 2014. doi: 10.1016/j.precisioneng.2014.02.013.
- Rich Caruana, Alexandru Niculescu-Mizil, Geoff Crew, and Alex Ksikes. Ensemble selection from libraries of models. In *Proceedings of the Twenty-First International Conference on Machine Learning (ICML ’04)*, page 18. ACM, 2004. doi: 10.1145/1015330.1015432.
- Arjun Chandra and Xin Yao. Ensemble learning using multi-objective evolutionary algorithms. *Journal of Mathematical Modelling and Algorithms*, 5(4):417–445, 2006. doi: 10.1007/s10852-005-9020-3.
- Lingjiao Chen, Matei Zaharia, and James Zou. FrugalML: How to use ML prediction APIs more accurately and cheaply. arXiv preprint arXiv:2006.07512, 2020. arXiv:2006.07512; commonly cited as NeurIPS 2020, venue not recorded in the arXiv metadata.
- Mayee F. Chen, Michael Y. Hu, Nicholas Lourie, Kyunghyun Cho, and Christopher Ré. Aioli: A Unified Optimization Framework for Language Model Data Mixing, 2025. Preprint; arXiv comment states ICLR 2025 camera ready.
- Tinkle Chugh, Yaochu Jin, Kaisa Miettinen, Jussi Hakanen, and Karthik Sindhya. A surrogate-assisted reference vector guided evolutionary algorithm for computationally expensive many-objective optimization. *IEEE Transactions on Evolutionary Computation*, 22(1):129–142, 2018. doi: 10.1109/TEVC.2016.2622301.

- Tinkle Chugh, Karthik Sindhya, Jussi Hakanen, and Kaisa Miettinen. A survey on handling computationally expensive multiobjective optimization problems with evolutionary algorithms. *Soft Computing*, 23(9):3137–3166, 2019. doi: 10.1007/s00500-017-2965-0.
- John A. Cornell. *Experiments with Mixtures: Designs, Models, and the Analysis of Mixture Data*. Wiley Series in Probability and Statistics. Wiley, Hoboken, NJ, 2002. doi: 10.1002/9781118204221.
- Danielle Martins Duarte Costa, Tarcísio Gonçalves Brito, Anderson Paulo de Paiva, Rafael Coradi Leme, and Pedro Paulo Balestrassi. A normal boundary intersection with multivariate mean square error approach for dry end milling process optimization of the AISI 1045 steel. *Journal of Cleaner Production*, 135:1658–1672, 2016. doi: 10.1016/j.jclepro.2016.01.062.
- Indraneel Das and J. E. Dennis. Normal-Boundary Intersection: A new method for generating the Pareto surface in nonlinear multicriteria optimization problems. *SIAM Journal on Optimization*, 8(3):631–657, 1998. doi: 10.1137/S1052623496307510.
- Fabício Alves de Almeida, Ana Carolina Oliveira Santos, Anderson Paulo de Paiva, Guilherme Ferreira Gomes, and José Henrique de Freitas Gomes. Multivariate Taguchi loss function optimization based on principal components analysis and normal boundary intersection. *Engineering with Computers*, 38(2):1627–1643, 2022. doi: 10.1007/s00366-020-01122-8.
- Taynara Incerti de Paula, Guilherme Ferreira Gomes, José Henrique de Freitas Gomes, and Anderson Paulo de Paiva. A mixture design of experiments approach for genetic algorithm tuning applied to multi-objective optimization. In *Optimization of Complex Systems: Theory, Models, Algorithms and Applications*, Advances in Intelligent Systems and Computing, pages 600–610. Springer International Publishing, 2019. ISBN 978-3-030-21803-4. doi: 10.1007/978-3-030-21803-4_60. Crossref issued 2019-06-15; Crossref published-print 2020.
- Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, 6(2):182–197, 2002. doi: 10.1109/4235.996017.
- Kalyanmoy Deb, Kaisa Miettinen, and Shamik Chaudhuri. Toward an estimation of nadir objective vector using a hybrid of evolutionary and local search approaches. *IEEE Transactions on Evolutionary Computation*, 14(6):821–841, 2010. doi: 10.1109/TEVC.2010.2041667.
- Kalyanmoy Deb, Rayan Hussein, Proteek Chandan Roy, and Gregorio Toscano-Pulido. A taxonomy for metamodeling frameworks for evolutionary multiobjective optimization. *IEEE Transactions on Evolutionary Computation*, 23(1):104–116, 2019. doi: 10.1109/TEVC.2018.2828091.
- Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7(1):1–30, 2006.
- Thomas G. Dietterich. Ensemble methods in machine learning. In *Multiple Classifier Systems (MCS 2000)*, *Lecture Notes in Computer Science*, pages 1–15. Springer, 2000. doi: 10.1007/3-540-45014-9_1.
- Eulanda M. Dos Santos, Robert Sabourin, and Patrick Maupin. A dynamic overproduce-and-choose strategy for the selection of classifier ensembles. *Pattern Recognition*, 41(10):2993–3009, 2008. doi: 10.1016/j.patcog.2008.03.027.
- Asif Ekbal and Sriparna Saha. A multiobjective simulated annealing approach for classifier ensemble: Named entity recognition in Indian languages as case studies. *Expert Systems with Applications*, 38(12):14760–14772, 2011. doi: 10.1016/j.eswa.2011.05.004.

- Asif Ekbal and Sriparna Saha. Combining feature selection and classifier ensemble using a multiobjective simulated annealing approach: Application to named entity recognition. *Soft Computing*, 17(1):1–16, 2013. doi: 10.1007/s00500-012-0885-6.
- Nick Erickson, Jonas Mueller, Alexander Shirkov, Hang Zhang, Pedro Larroy, Mu Li, and Alexander Smola. AutoGluon-Tabular: Robust and accurate AutoML for structured data. arXiv preprint arXiv:2003.06505, 2020. arXiv:2003.06505.
- Matthias Feurer, Katharina Eggensperger, Stefan Falkner, Marius Lindauer, and Frank Hutter. Auto-Sklearn 2.0: Hands-free AutoML via meta-learning. *Journal of Machine Learning Research*, 23(261), 2022. arXiv:2007.04074.
- Giorgio Fumera and Fabio Roli. A theoretical and experimental analysis of linear combiners for multiple classifier systems. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(6):942–956, 2005. doi: 10.1109/TPAMI.2005.109.
- José Galván, Gracia Sánchez, and Fernando Jiménez. Simultaneous multi-objective evolutionary optimization of heterogeneous ensembles, learner-specific feature subsets, and aggregation weights. *Algorithms*, 19(8):681, 2026. doi: 10.3390/a19080681.
- Peter Anton Gellerich and Jens-Peter Majschak. A design of experiments based on the Normal-boundary-intersection method to identify optimum machine settings in manufacturing processes. *PAMM*, 23(3):e202300009, 2023. doi: 10.1002/pamm.202300009.
- Jashwant Raj Gunasekaran, Cyan Subhra Mishra, Prashanth Thinakaran, Bikash Sharma, Mahmut Taylan Kandemir, and Chita R. Das. Cocktail: A multidimensional optimization for model serving in cloud. In *19th USENIX Symposium on Networked Systems Design and Implementation (NSDI 22)*, pages 1041–1057, Renton, WA, 2022. USENIX Association. ISBN 978-1-939133-27-4. Preprint: arXiv:2106.05345.
- Lars Kai Hansen and Peter Salamon. Neural network ensembles. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12(10):993–1001, 1990. doi: 10.1109/34.58871.
- Alaa Hassan, Reda Mohamed, Mohamed Abdel-Basset, and Mohamed Abouhawwash. A weighted voting-based ensemble classifier based on manta ray foraging optimizer for cyber-attack detection in IoT environments: A comparative study. *Evolutionary Intelligence*, 19(1): 18, 2026. doi: 10.1007/s12065-025-01126-7.
- Linjun He, Hisao Ishibuchi, Anupam Trivedi, Handing Wang, Yang Nan, and Dipti Srinivasan. A survey of normalization methods in multiobjective evolutionary algorithms. *IEEE Transactions on Evolutionary Computation*, 25(6):1028–1048, 2021. doi: 10.1109/TEVC.2021.3076514.
- Markus Herrmann-Wicklmayr and Kathrin Flaßkamp. Non-extreme individual minima for improved Pareto front sampling efficiency and decision-making. In *Computational Science – ICCS 2026 Workshops, Lecture Notes in Computer Science*, pages 271–284. Springer Nature Switzerland, 2026. doi: 10.1007/978-3-032-29909-3_20. Preprint arXiv:2602.21883.
- Heinz Isermann and Ralph E. Steuer. Computational experience concerning payoff tables and minimum criterion values over the efficient set. *European Journal of Operational Research*, 33(1):91–97, 1988. doi: 10.1016/0377-2217(88)90257-3.
- Hisao Ishibuchi, Hiroyuki Masuda, Yuki Tanigaki, and Yusuke Nojima. Modified distance calculation in generational distance and inverted generational distance. In *Evolutionary Multi-Criterion Optimization (EMO 2015), Lecture Notes in Computer Science*, pages 110–125. Springer, 2015. doi: 10.1007/978-3-319-15892-1_8.

- Junzhong Ji and Junwei Li. Tri-objective optimization-based cascade ensemble pruning for deep forest. *Pattern Recognition*, 143:109744, 2023. doi: 10.1016/j.patcog.2023.109744.
- Yaochu Jin. Surrogate-assisted evolutionary computation: Recent advances and future challenges. *Swarm and Evolutionary Computation*, 1(2):61–70, 2011. doi: 10.1016/j.swevo.2011.05.001.
- Yaochu Jin and Bernhard Sendhoff. Pareto-based multiobjective machine learning: An overview and case studies. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 38(3):397–415, 2008. doi: 10.1109/TSMCC.2008.919172.
- Andrew T. Karl, Sean Essex, James Wisnowski, and Heath Rushing. A Workflow for Lipid Nanoparticle (LNP) Formulation Optimization using Designed Mixture-Process Experiments and Self-Validated Ensemble Models (SVEM). *Journal of Visualized Experiments*, (198), 2023. doi: 10.3791/65200. Also available as arXiv:2212.11264.
- Josef Kittler, Mohamad Hatef, Robert P. W. Duin, and Jiri Matas. On combining classifiers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(3):226–239, 1998. doi: 10.1109/34.667881.
- J. Knowles. ParEGO: A hybrid algorithm with on-line landscape approximation for expensive multiobjective optimization problems. *IEEE Transactions on Evolutionary Computation*, 10(1):50–66, 2006. doi: 10.1109/TEVC.2005.851274.
- Ludmila I. Kuncheva. *Combining Pattern Classifiers: Methods and Algorithms*. Wiley, Hoboken, NJ, 2nd edition, 2014. doi: 10.1002/9781118914564. Crossref reports edition-number 1; the 2nd-edition statement is from the publisher listing and was not confirmed in the fetched record.
- Youngseok Kwon, Kwanghyun Lee, and Dongju Lee. Ensemble Prediction Model Using Mixture Design of Experiments. *Journal of Society of Korea Industrial and Systems Engineering*, 47(4):161–170, 2024. doi: 10.11627/jksie.2024.47.4.161.
- Gustavo dos Santos Leal, Estevão Luiz Romão, Daniel Leal de Paula Esteves dos Reis, Pedro Paulo Balestrassi, and Anderson Paulo de Paiva. A multiobjective portfolio optimization for energy assets using D-Optimal design and mixture design of experiments. *Production*, 32:e20210119, 2022. doi: 10.1590/0103-6513.20210119.
- Gustavo S. Leal, Lupércio F. Bessegato, Yasmin S. M. Xavier, Farid Melgani, and Pedro Paulo Balestrassi. A design-of-experiments-based approach for efficient estimation of bimodal Gaussian mixture weights. *IEEE Access*, 13:168322–168334, 2025. doi: 10.1109/ACCESS.2025.3614023.
- Michael LeBlanc and Robert Tibshirani. Combining estimates in regression and classification. *Journal of the American Statistical Association*, 91(436):1641–1650, 1996. doi: 10.1080/01621459.1996.10476733.
- Erin LeDell, Mark J. van der Laan, and Maya Petersen. AUC-maximizing ensembles through metalearning. *The International Journal of Biostatistics*, 12(1):203–218, 2016. doi: 10.1515/ijb-2015-0035.
- Qian Liu, Xiaosen Zheng, Niklas Muennighoff, Guangtao Zeng, Longxu Dou, Tianyu Pang, Jing Jiang, and Min Lin. RegMix: Data Mixture as Regression for Language Model Pre-training, 2025. Preprint; arXiv comment states ICLR 2025.
- Luiz Gustavo Dias Lopes, Tarcísio Gonçalves Brito, Anderson Paulo Paiva, Rogério Santana Peruchi, and Pedro Paulo Balestrassi. Robust parameter optimization based on multivariate

- normal boundary intersection. *Computers & Industrial Engineering*, 93:55–66, 2016. doi: 10.1016/j.cie.2015.12.023.
- Jannis Maier and Lennart Purucker. HAPEns: Hardware-aware post-hoc ensembling for tabular data. arXiv preprint arXiv:2603.10582, 2026. arXiv:2603.10582v2.
- Jannis Maier, Felix Möller, and Lennart Purucker. Hardware aware ensemble selection for balancing predictive accuracy and cost. In *Third International Conference on Automated Machine Learning (AutoML 2024), Workshop Track*, 2024. arXiv:2408.02280.
- R. R. A. Mendes, A. P. Paiva, R. S. Peruchi, P. P. Balestrassi, R. C. Leme, and M. B. Silva. Multiobjective portfolio optimization of ARMA–GARCH time series based on experimental designs. *Computers & Operations Research*, 66:434–444, 2016. doi: 10.1016/j.cor.2015.05.001.
- A. Messac, A. Ismail-Yahaya, and C. A. Mattson. The normalized normal constraint method for generating the Pareto frontier. *Structural and Multidisciplinary Optimization*, 25(2):86–98, 2003. doi: 10.1007/s00158-002-0276-1.
- Larissa Ferreira Rodrigues Moreira, Rodrigo Moreira, and Leonardo Gabriel Ferreira Rodrigues. Energy-aware ensemble learning for coffee leaf disease classification. arXiv preprint arXiv:2601.12109, 2026. arXiv:2601.12109.
- M. O. Moreira, P. P. Balestrassi, A. P. Paiva, P. F. Ribeiro, and B. D. Bonatto. Design of experiments using artificial neural network ensemble for photovoltaic generation forecasting. *Renewable and Sustainable Energy Reviews*, 135:110450, 2021. doi: 10.1016/j.rser.2020.110450.
- Raymond H. Myers, Douglas C. Montgomery, and Christine M. Anderson-Cook. *Response Surface Methodology: Process and Product Optimization Using Designed Experiments*. Wiley Series in Probability and Statistics. John Wiley & Sons, Hoboken, NJ, 4th edition, 2016. ISBN 9781118916018. Authors, edition (“4, illustrated”), series, publisher, year, ISBN and extent all confirmed by the cached Google Books record for ISBN 9781118916018, and corroborated by OpenLibrary (title, subtitle, 2016, 856 pp.) and the Journal of Quality Technology review of the 4th edition (Jensen, 2017, 49(2):186–188). The book has no Crossref or OpenAlex record.
- Claude Nadeau and Yoshua Bengio. Inference for the generalization error. *Machine Learning*, 52(3):239–281, 2003. doi: 10.1023/A:1024068626366.
- Feng Nan and Venkatesh Saligrama. Adaptive classification for prediction under a budget. arXiv preprint arXiv:1705.10194, 2017. arXiv:1705.10194; commonly cited as NIPS 2017, venue not recorded in the arXiv metadata.
- Feng Nan, Joseph Wang, and Venkatesh Saligrama. Feature-budgeted random forest. arXiv preprint arXiv:1502.05925, 2015. arXiv:1502.05925; commonly cited as ICML 2015, venue not recorded in the arXiv metadata.
- Fabiano Luiz Naves, Taynara Incerti de Paula, Pedro Paulo Balestrassi, Washington Luis Moreira Braga, Rapinder Singh Sawhney, and Anderson Paulo de Paiva. Multivariate Normal Boundary Intersection based on rotated factor scores: A multiobjective optimization method for methyl orange treatment. *Journal of Cleaner Production*, 143:413–439, 2017. doi: 10.1016/j.jclepro.2016.12.092.
- Aytuğ Onan, Serdar Korukoğlu, and Hasan Bulut. A multiobjective weighted voting ensemble classifier based on differential evolution algorithm for text sentiment classification. *Expert Systems with Applications*, 62:1–16, 2016. doi: 10.1016/j.eswa.2016.06.005.

- Matheus Costa Pereira, Caio Tertuliano Ribeiro, Ronã Rinston Amaury Mendes, Paulo Henrique da Silva Campos, and Anderson Paulo de Paiva. A hybrid multivariate normal boundary intersection approach with post-optimization assisted by mixture design of experiments. *Engineering Applications of Artificial Intelligence*, 162:112510, 2025. doi: 10.1016/j.engappai.2025.112510.
- Gregory F. Piepel. Measuring Component Effects in Constrained Mixture Experiments. *Technometrics*, 24(1):29–39, 1982. doi: 10.1080/00401706.1982.10487706.
- Wolfgang Ponweiser, Tobias Wagner, Dirk Biermann, and Markus Vincze. Multiobjective optimization on a limited budget of evaluations using model-assisted S-metric selection. In *Parallel Problem Solving from Nature – PPSN X, Lecture Notes in Computer Science*, pages 784–794. Springer Berlin Heidelberg, 2008. doi: 10.1007/978-3-540-87700-4_78.
- Chao Qian, Yang Yu, and Zhi-Hua Zhou. Pareto ensemble pruning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 29(1), 2015. doi: 10.1609/aaai.v29i1.9579.
- Kimia Rezaei, Hamed Agahi, and Azar Mahmoodzadeh. Multi-objective differential evolution-based ensemble method for brain tumour diagnosis. *IET Image Processing*, 13(9):1421–1430, 2019. doi: 10.1049/iet-ipr.2018.6377.
- Caio Tertuliano Ribeiro. Meta-otimização dos hiperparâmetros do algoritmo XGBoost para classificação binária: uma integração entre planejamento de experimentos e o método da interseção normal à fronteira. Master’s thesis, Universidade Federal de Itajubá, Programa de Pós-Graduação em Engenharia de Produção, Itajubá, Brazil, February 2026. Advisor: Anderson Paulo de Paiva. 153 f. Available at <https://repositorio.unifei.edu.br/jspui/handle/123456789/4372>.
- Victor Henrique Alves Ribeiro and Gilberto Reynoso-Meza. Ensemble learning by means of a multi-objective optimization design approach for dealing with imbalanced data sets. *Expert Systems with Applications*, 147:113232, 2020. doi: 10.1016/j.eswa.2020.113232.
- Franco B. Rocha, Paulo Rotella, Pedro Paulo Balestrassi, Farid Melgani, and Antonio C. Zambroni De Souza. Optimizing neural network ensemble for short-term load forecasting: A case study of a Brazilian electrical distribution system. *IEEE Access*, 13:207903–207915, 2025. doi: 10.1109/ACCESS.2025.3641712.
- Henry Scheffé. Experiments with mixtures. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 20(2):344–360, 1958. doi: 10.1111/j.2517-6161.1958.tb00299.x.
- Henry Scheffé. The simplex-centroid design for experiments with mixtures. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 25(2):235–251, 1963. doi: 10.1111/j.2517-6161.1963.tb00506.x.
- Mohsen Shahhosseini, Guiping Hu, and Hieu Pham. Optimizing ensemble weights and hyperparameters of machine learning models for regression problems. *Machine Learning with Applications*, 7:100251, 2022. doi: 10.1016/j.mlwa.2022.100251.
- Mohammad Tabatabaei, Jussi Hakanen, Markus Hartikainen, Kaisa Miettinen, and Karthik Sindhya. A survey on handling computationally expensive multiobjective optimization problems using surrogates: Non-nature inspired methods. *Structural and Multidisciplinary Optimization*, 52(1):1–25, 2015. doi: 10.1007/s00158-015-1226-z.
- Anvith Thudi, Evianne Rovers, Yangjun Ruan, Tristan Thrush, and Chris J. Maddison. MixMin: Finding Data Mixtures via Convex Minimization, 2025. Preprint; arXiv comment states Proceedings of the 42nd International Conference on Machine Learning (ICML 2025).

- Kirill Trapeznikov, Venkatesh Saligrama, and David Castañón. Multi-stage classifier design. *Machine Learning*, 92(2-3):479–502, 2013. doi: 10.1007/s10994-013-5349-4.
- Peter D. Turney. Types of cost in inductive concept learning. In *Workshop on Cost-Sensitive Learning at the Seventeenth International Conference on Machine Learning (ICML 2000)*, Stanford University, California, pages 15–21, 2000. arXiv:cs/0212034.
- Mark J. van der Laan, Eric C. Polley, and Alan E. Hubbard. Super learner. *Statistical Applications in Genetics and Molecular Biology*, 6(1):Article 25, 2007. doi: 10.2202/1544-6115.1309.
- Jasper A. Vrugt, Martyn P. Clark, Cees G. H. Diks, Qingyun Duan, and Bruce A. Robinson. Multi-objective calibration of forecast ensembles using Bayesian model averaging. *Geophysical Research Letters*, 33(19):2006GL027126, 2006. doi: 10.1029/2006GL027126. Crossref deposits no page range (article number 2006GL027126) and misspells the fourth author as “Qinyun Duan”; the cached OpenAlex record gives “Qingyun Duan”, used here.
- Xiaofang Wang, Dan Kondratyuk, Eric Christiansen, Kris M. Kitani, Yair Alon, and Elad Eban. Wisdom of committees: An overlooked approach to faster and more accurate models. In *International Conference on Learning Representations (ICLR 2022)*, 2022. arXiv:2012.01988.
- David H. Wolpert. Stacked generalization. *Neural Networks*, 5(2):241–259, 1992. doi: 10.1016/S0893-6080(05)80023-1.
- Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and Adams Wei Yu. DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining, 2023. Preprint; arXiv comment states NeurIPS 2023.
- Beicheng Xu, Wei Liu, Keyao Ding, Yupeng Lu, and Bin Cui. PSEO: Optimizing post-hoc stacking ensemble through hyperparameter tuning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 40(32):27188–27196, 2026. doi: 10.1609/aaai.v40i32.39934.
- Zhixiang Xu, Kilian Weinberger, and Olivier Chapelle. The greedy miser: Learning under test-time budgets. In *Proceedings of the 29th International Conference on Machine Learning (ICML 2012)*, 2012. arXiv:1206.6451.
- Zhixiang Xu, Matt J. Kusner, Kilian Q. Weinberger, and Minmin Chen. Cost-sensitive tree of classifiers. arXiv preprint arXiv:1210.2771, 2013. arXiv:1210.2771; commonly cited as ICML 2013, venue not recorded in the arXiv metadata.
- Yuling Yao, Aki Vehtari, Daniel Simpson, and Andrew Gelman. Using stacking to average Bayesian predictive distributions (with discussion). *Bayesian Analysis*, 13(3), 2018. doi: 10.1214/17-BA1091.
- Jiasheng Ye, Peiju Liu, Tianxiang Sun, Jun Zhan, Yunhua Zhou, and Xipeng Qiu. Data Mixing Laws: Optimizing Data Mixtures by Predicting Language Modeling Performance, 2025. Preprint; arXiv comment states accepted by ICLR 2025.
- Li Zhang and Wei-Da Zhou. Sparse ensembles using weighted combination methods based on linear programming. *Pattern Recognition*, 44(1):97–106, 2011. doi: 10.1016/j.patcog.2010.07.021.
- Qingfu Zhang, Wudong Liu, Edward Tsang, and Botond Virginas. Expensive multiobjective optimization by MOEA/D with Gaussian process model. *IEEE Transactions on Evolutionary Computation*, 14(3):456–474, 2010. doi: 10.1109/TEVC.2009.2033671.

- Jiaqi Zhao, Licheng Jiao, Shixiong Xia, Vitor Basto Fernandes, Iryna Yevseyeva, Yong Zhou, and Michael T. M. Emmerich. Multiobjective sparse ensemble learning by means of evolutionary algorithms. *Decision Support Systems*, 111:86–100, 2018. doi: 10.1016/j.dss.2018.05.003.
- Zhi-Hua Zhou, Jianxin Wu, and Wei Tang. Ensembling neural networks: Many could be better than all. *Artificial Intelligence*, 137(1-2):239–263, 2002. doi: 10.1016/S0004-3702(02)00190-X.